When Does Sparsification Help Community Detection? Separating Genuine Gains from Degree-Mechanical Artifacts

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Abstract—Graph sparsification is increasingly reported to improve community detection rather than simply accelerate it. We show that, for degree-based sparsification, the standard evidence for such improvements is largely an artifact of evaluation methodology, and we characterize the narrow conditions under which genuine gains exist. We first develop a fixed-partition theory of degree-based edge sampling: when intra-community edges receive higher DSpar scores than inter-community edges (positive separation, $\delta > 0$), score-proportional sampling provably removes inter-community edges preferentially and increases the intra-community edge fraction. We then show empirically that this mechanism operates independently of community structure: on 17 real-world networks, degree-preserving configuration-model nulls reproduce both $\delta > 0$ and fixed-partition modularity gains on every network, typically exceeding the real graph (median ratio 1.24), so neither signal certifies community structure; on the contrary, real structure typically weakens both. We further identify three evaluation artifacts that generate apparent improvements: (i) comparing modularity across different graphs: partitions optimized on sparsified graphs lose 0.004–0.19 modularity when evaluated on the original network, on all 15 networks tested; (ii) partition granularity: apparent ground-truth-recovery gains vanish against resolution-matched controls run on the unsparsified graph, on every labeled dataset tested; and (iii) unmatched computation: gains attributed to sparsification-seeded optimization disappear against runtime-matched restart baselines on 13 of 15 networks. After controlling for all three, a genuine benefit survives on 2 of 15 networks: seeding Leiden with the sparsified-graph partition and refining on the original graph beats runtime-matched baselines robustly on one network (+0.008 modularity) and by a small margin confined to the mildest sparsification on a second (+0.003); no structural statistic we tested predicts where such gains occur. Our results yield a practical evaluation protocol (null-model, resolution-matched, and runtime-matched controls) for any claim that preprocessing improves community detection.

Index Terms—Graph sparsification, community detection, modularity maximization, null models, evaluation methodology, degree heterogeneity.

1 Introduction

Community detection, the task of identifying densely connected groups of nodes, is a fundamental problem in network science [1], [2]. As networks scale to millions of nodes and edges, graph sparsification, the removal of edges while approximately preserving structural properties, has become a standard preprocessing step, classically justified by computational savings alone [3], [4]. Recently, a more ambitious claim has emerged: that sparsification can improve downstream community detection rather than merely accelerate it. Similarity-based sparsification has been reported to sharpen cluster structure [5]; degree-based schemes such as DSpar [6] preferentially remove edges between high-degree hubs, which are intuitively the edges most likely to blur community boundaries [7], [8]; and, in an earlier version of this work, we ourselves reported consistent modularity improvements from DSpar preprocessing across a broad collection of real-world networks (unpublished manuscript).

This paper subjects that class of claims to controlled evaluation, and the result is largely negative: for degree-based sparsification, the standard evidence of improved community detection is dominated by measurement artifacts rather than genuine structural gains. At the same time, the underlying mechanism is real and provable, and a small genuine benefit survives the strictest controls we could devise, on 2 of 15 networks. Determining which part is real and which is artifact, and providing the controls that separate them, is the contribution of this paper.

1.0.0.1 The mechanism is real.: Degree-based sparsification assigns each edge (u, v) a score $s(u, v) = 1/d_u + 1/d_v$ and retains edges with probability increasing in this score, so edges between high-degree nodes are removed preferentially. When intra-community edges receive higher scores on average than inter-community edges, a condition we call DSpar separation ($\delta > 0$), we prove that sparsification removes inter-community edges preferentially and increases the intra-community edge fraction of any fixed partition (Section 3). These results hold under both deterministic edge weighting and randomized edge removal for score-proportional sampling; Sections 4 and 5 delimit how far the samplers used in practice depart from that regime, and what survives the departure.

1.0.0.2 The evidence is confounded.: The mechanism, however, does not imply improved community detection, because the standard ways of measuring improvement are confounded by three distinct artifacts. First, modularity is not comparable across graphs: sparsification changes the edge set

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and the configuration null model simultaneously, and sparser graphs support higher modularity for purely combinatorial reasons [9]. When partitions optimized on sparsified graphs are evaluated on the original network, they lose 0.004–0.19 modularity, on every one of the 15 networks we test. Second, partition granularity: sparsification fragments graphs, detection algorithms return finer partitions, and popular agreement measures reward finer partitions against small ground-truth communities [10]. A resolution-matched control, meaning the same detection algorithm run on the unsparsified graph and tuned to the same number of clusters, matches or beats sparsification on every labeled dataset we test. Third, unmatched computation: sparsify-cluster-refine pipelines spend more optimization effort than the single-run baselines they are compared against; when baselines are granted equal wall-clock time via restarts, most apparent gains disappear.

1.0.0.3 The certifying signals do not certify.: More fundamentally, we show that the structural signals used to certify the mechanism are not evidence of community structure. On all 17 networks in our suite, degree-preserving configuration-model rewiring, which destroys community structure while retaining the degree sequence, reproduces both $\delta > 0$ and positive fixed-partition modularity change under sparsification, typically at larger magnitude than the real graph (median ratio 1.24). Hub-bridging behaves identically: the degree-product ratio between inter- and intra-community edges exceeds one on every rewired null, while it is erratic on the real graphs. Both quantities are properties of heavy-tailed degree sequences interacting with modularity optimization, not of community structure; neither can distinguish a real network from its own configuration model.

1.0.0.4 What survives.: After all three controls, one genuine effect remains: seeding the Leiden algorithm [11] with the partition found on the sparsified graph and refining on the original graph beats runtime-matched restart baselines on 2 of 15 networks: email-Enron, robust across every configuration tested (about four baseline standard deviations at the main setting), and com-Youtube, a smaller gain confined to the mildest calibrated sparsification. None of the structural statistics we tested (δ, its null-corrected excess, hub-bridging strength, degree heterogeneity, graph size) predicts where these gains occur. The strongest candidate correlation appeared statistically significant over 14 networks ($r = 0.69$, $p = 0.006$) and was eliminated by the fifteenth ($r = 0.12$), a cautionary illustration of small-sample correlation claims in this literature, including one made in the earlier version of this work.

Contributions.

- A fixed-partition theory of degree-based sparsification (Section 3): DSpar separation provably yields preferential inter-community edge removal and intra-community fraction gains, under a corrected sampler formulation whose realized retention matches its nominal rate exactly.
- A null-model characterization showing that DSpar separation and fixed-partition modularity gains are reproduced, and usually amplified, by degree-preserving configuration models on 17/17 real networks, and are therefore not evidence of community structure; indeed, real structure systematically weakens both signals relative to the null.
- A three-control evaluation protocol for preprocessing claims in community detection: score partitions on the original graph (not the preprocessed one), compare against resolution-matched controls, and grant baselines matched computation. Each control is paired with the artifact it exposes, with open implementations.
- A boundary characterization: genuine runtime-matched gains exist on 2 of 15 networks (one robust, one small and regime-confined), and no tested structural statistic predicts them: an existence proof, an open problem, and a demonstration that predictor claims in this regime require far larger samples.
- A sampler audit showing that the nominal retention parameter α of published DSpar-style samplers differs systematically from realized edge retention (by more than a factor of two), with a calibrated sampler that closes the gap.

The remainder of the paper is organized as follows. Section 2 reviews background and related work. Section 3 develops the fixed-partition theory. Section 4 audits the samplers. Sections 5 and 6 present the mechanism verification with its null-model controls and the three evaluation artifacts. Section 7 characterizes the boundary where genuine gains exist, Section 8 reports runtime behavior, and Section 9 concludes.

2 Background

We review three foundational concepts, modularity-based community detection, spectral graph sparsification, and degree-based sparsification, then establish their theoretical connections.

Community detection seeks to partition a network into groups with dense internal connections and sparse external connections [12]. Among the many formulations of this problem, modularity maximization [13], [14] remains one of the most widely used.

For a graph $G = (V, E)$ with n nodes and m edges, the modularity Q of a partition is defined as:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{1}$$

where A_{ij} is the adjacency matrix, k_i is the degree of node i, c_i denotes the community assignment of node i, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same community.[1] The term $k_i k_j / 2m$ represents the expected number of edges between nodes i and j under the configuration model, a null model that preserves the degree sequence but places edges uniformly at random [15], [16]. Modularity thus measures the deviation of intra-community edge density from this null expectation.

Equation (1) can be rewritten as a sum over communities:

$$Q = \sum_{c=1}^{C} \left[\frac{L_c}{m} - \left(\frac{d_c}{2m} \right)^2 \right] \tag{2}$$

where C is the number of communities, L_c is the number of edges internal to community c, and $d_c = \sum_{i \in c} k_i$ is the sum of

1. The Kronecker delta always appears with its arguments; a bare δ denotes the DSpar separation gap introduced in Section 3.

degrees of nodes in community c [14]. The first term rewards internal edges, while the second term penalizes communities whose nodes have high total degree.

This formulation reveals a key asymmetry: inter-community edges contribute to d_c (they increase node degrees) but not to L_c (they are not internal). Consequently, for a fixed partition, downweighting or removing inter-community edges tends to increase the intra-community fraction relative to the degree-based null model, although the net modularity change also depends on how node degrees and total edge count change after sparsification.

Maximizing modularity is NP-hard [17], but effective heuristics exist. The Louvain algorithm [18] uses greedy local moves with hierarchical aggregation. The Leiden algorithm [11] refines this approach by guaranteeing well-connected communities and improving partition quality. We use the Leiden algorithm throughout this work.

Graph sparsification constructs a sparse subgraph H that approximates properties of the original graph G. Spectral sparsification, introduced by Spielman and Teng [19] and refined by Spielman and Srivastava [3], preserves the spectral properties of the graph Laplacian.

The combinatorial Laplacian is defined as $L = D - A$, where D is the diagonal degree matrix. For any vector $x \in \mathbb{R}^n$, the Laplacian quadratic form equals:

$$x^\top Lx = \sum_{(i,j)\in E} (x_i - x_j)^2 \tag{3}$$

This quantity measures the "smoothness" of x over the graph and appears in spectral clustering [20], random walks [21], and electrical network theory [22].

A graph H is a $(1 \pm \epsilon)$-spectral sparsifier of G if for all $x \in \mathbb{R}^n$:

$$(1-\epsilon)\cdot x^\top L_G x \le x^\top L_H x \le (1+\epsilon)\cdot x^\top L_G x \tag{4}$$

Spielman and Srivastava [3] proved that sampling $O(n \log n/\epsilon^2)$ edges with probability proportional to their effective resistance yields a spectral sparsifier with high probability.

The effective resistance between nodes u and v is defined as:

$$R_{\text{eff}}(u,v) = (\chi_u - \chi_v)^\top L^+ (\chi_u - \chi_v) \tag{5}$$

where L^+ is the Moore-Penrose pseudoinverse of the Laplacian and χ_u is the indicator vector for node u. This quantity has an intuitive interpretation: if the graph is viewed as an electrical network with unit resistors on each edge, $R_{\text{eff}}(u,v)$ is the voltage difference induced by injecting one unit of current at u and extracting it at v [22].

Edges with high effective resistance are critical for connectivity, removing them significantly alters the graph's spectral properties. Edges with low effective resistance are often embedded in regions with many alternative paths and are therefore less critical to preserving global spectral properties.

Computing effective resistances requires $O(n^3)$ time for matrix inversion, or $O(m + n \log n)$ time per resistance using Laplacian solvers [23]. While approximate effective-resistance sampling can be implemented in near-linear time using advanced Laplacian solvers, it remains substantially more complex than degree-based methods in practice. (For large networks, this cost can exceed that of community detection itself, motivating faster approximations.)

Classical results in electrical network theory show that effective resistance is closely related to vertex degrees and graph connectivity; in particular, degree-based bounds follow from Rayleigh monotonicity and appear in the work of Doyle and Snell [22] and later in the spectral sparsification framework of Spielman and Srivastava [3]. Lovász [21] provides a broader survey perspective, connecting effective resistance to random walks and global spectral properties of the graph Laplacian. These results imply that edges bridging weakly connected regions typically exhibit higher effective resistance than edges embedded within dense subgraphs.

DSpar [6] leverages this intuition by assigning each edge $e = (u,v)$ the score

$$s(u,v) = \frac{1}{d_u} + \frac{1}{d_v}, \tag{6}$$

where d_u and d_v denote the endpoint degrees, and by sampling edges with probability increasing in $s(u,v)$. The DSpar score has an intuitive interpretation:

- High score: edge connects low-degree peripheral nodes, likely important for local connectivity
- Low score: edge connects high-degree hub nodes, likely redundant due to alternative paths

DSpar requires only degree information, computable in $O(m)$ time via a single pass over the edge list. In contrast, effective-resistance sampling requires solving Laplacian systems; even with near-linear-time solvers [23] it remains substantially more complex in practice, making degree-based scores attractive for networks with millions of edges.

Related work

Beyond spectral sparsification, several lines of work remove edges to aid analysis. Satuluri et al. [5] sparsify by local structural similarity explicitly to improve scalable clustering, and backbone-extraction methods retain statistically significant edges in weighted networks [24], [25]. Our results complement this literature by identifying which evaluation designs can and cannot certify such improvements. A second line concerns the behavior of modularity itself: random graphs without community structure exhibit substantial modularity [9], and modularity maximization has a resolution limit tied to the total edge count [26]. Both facts bear directly on comparisons of scores across graphs of different density, which sparsification pipelines perform implicitly. For comparing partitions against reference labels, information-theoretic measures require correction for chance, since uncorrected NMI is biased toward fine partitions [10], and metadata labels are imperfect proxies for structural communities [27], [28]. Finally, post-processing methods that repair poorly connected clusters [29], [30], [31] address a related failure mode of modularity pipelines from the opposite direction; connecting the pre- and post-processing perspectives remains open.

3 Theoretical Connections

We now clarify the relationships between modularity-based community detection, spectral graph methods, and degree-based sparsification, and identify a shared structural bias common to all three. Although real networks typically contain multiple communities, many spectral relationships are most transparent in the bipartition setting. We therefore use bipartition-based spectral relaxations to develop intuition, while all subsequent definitions, hypotheses, and results apply to arbitrary multi-community partitions.

Newman [32] showed that several widely used community detection objectives, including modularity maximization, statistical inference under the degree-corrected stochastic block model [33], and normalized-cut graph partitioning [34], admit closely related spectral relaxations. In each case, the discrete partitioning problem is relaxed to a continuous optimization whose solution is given by an eigenvector of a degree-normalized operator.

For normalized cuts and related random-walk formulations, this operator is the normalized adjacency matrix

$$\mathcal{A} = D^{-1/2} A D^{-1/2}, \tag{7}$$

which is similar to the random-walk transition matrix $D^{-1}A$. In the bipartition setting, the relaxed solution is determined by the second-largest eigenvector of $\mathcal{A}$ (or equivalently, the second-smallest eigenvector of the normalized Laplacian $I - \mathcal{A}$), with a discrete partition obtained by thresholding or taking signs. For multi-community detection, multiple leading eigenvectors are used and rounded via clustering. The modularity matrix B, defined by $B_{ij} = A_{ij} - k_i k_j / 2m$, plays a central role in spectral modularity maximization [14]. Newman [32] showed that, under a degree-weighted spectral relaxation and the orthogonality constraint $k^\top s = 0$, the generalized eigenproblem

$$Bs = \lambda D s \tag{8}$$

reduces to

$$As = \lambda D s. \tag{9}$$

After the similarity transform $u = D^{1/2} s$, this is equivalent to an eigenproblem involving the normalized adjacency matrix $\mathcal{A}$. Thus, although the underlying objectives differ, their commonly used spectral relaxations lead to closely related degree-normalized eigenproblems. Despite their different formulations, modularity maximization, normalized-cut methods, spectral sparsification, and degree-based sparsification all encode a common structural bias:

Edges connecting high-degree nodes are systematically downweighted or deemphasized under degree-corrected null models, degree-normalized operators, and resistance- or degree-based sparsification schemes.

This bias manifests differently across frameworks:

- Modularity: The null-model term $k_i k_j / 2m$ is large for high-degree node pairs, reducing their contribution to the modularity objective.
- Normalized cuts: Degree normalization in operators such as $\mathcal{A} = D^{-1/2} A D^{-1/2}$ reduces the influence of hubs on the relaxed partitioning objective.
- Spectral sparsification: Edges embedded in highly connected regions typically have low effective resistance and are more likely to be removed when constructing a spectral sparsifier.
- DSpar: The score $s(u, v) = 1/d_u + 1/d_v$ assigns low sampling probability to hub–hub edges, favoring retention of edges incident to low-degree nodes.

These mechanisms do not imply that hub–hub edges are universally uninformative, but rather that all four frameworks preferentially discount edges that are expected under degree-driven null models or dense connectivity.

3.1 The Hub–Bridge Hypothesis

The connections above identify a structural condition under which degree-based sparsification biases edge removal toward inter-community edges. We state it as a premise about a graph together with a partition; whether it reflects community structure or degree structure alone is an empirical question, answered in Section 5 with configuration-model controls.

Hypothesis 1 (Hub–Bridge Condition). A graph–partition pair $(G, \mathcal{C})$ exhibits hub-bridging if inter-community edges connect higher-degree endpoints on average than intra-community edges:

$$\mathbb{E}[d_u d_v \mid (u, v) \in E_{\text{inter}}] > \mathbb{E}[d_u d_v \mid (u, v) \in E_{\text{intra}}]. \tag{10}$$

Two complementary mechanisms can generate hub-bridging in real networks.

Probabilistic mechanism. In networks with heterogeneous degree distributions and finite community sizes, high-degree nodes exhaust available intra-community connections and must place surplus edges externally: a node of degree d in a community of n_c nodes has at least $d - n_c + 1$ external edges whenever $d \geq n_c$.

Functional mechanism. In social and information networks, hubs often play bridging roles across groups [7], [8], such as prolific authors linking research areas or influential users connecting communities. These functional roles naturally induce inter-community connectivity among high-degree nodes.

When the condition holds, degree-based sparsification tends to assign lower sampling probability to inter-community edges, since such edges typically connect high-degree endpoints. As a result, DSpar preferentially removes inter-community edges relative to intra-community edges, which can increase the intra-community edge fraction and, for a fixed partition, improve modularity.

The condition is unlikely to hold when:

1) the degree distribution is homogeneous (no hubs), or
2) inter-community edges do not preferentially involve high-degree nodes.

Whether hub-bridging holds depends on the partition used and, as Section 5.2 shows, is not by itself informative about community structure: three networks in our suite violate it under Leiden partitions while retaining $\delta > 0$, and every rewired null satisfies it.

3.2 Theoretical Analysis

Our theoretical results characterize how degree-based weighting and sparsification bias edge retention for a fixed partition. They make no claim that positive DSpar separation indicates community structure, and no claim about optimized detection outcomes: Section 5 shows that the premise $\delta > 0$ (and the resulting fixed-partition gains) arises equally on configuration-model graphs without communities, and Section 6 analyzes why fixed-partition gains must not be read as detection improvements. The theory is thus a quantitative account of a mechanism; its interpretation is settled empirically.

Definition 1 (DSpar Score). For an edge $e = (u, v)$ in graph $G = (V, E)$, the DSpar score is:

$$s(e) = \frac{1}{d_u} + \frac{1}{d_v} \tag{11}$$

where d_u, d_v denote the degrees of nodes u and v respectively.

Definition 2 (Edge Classification). Given a partition $\mathcal{C} = \{C_1, \ldots, C_k\}$ of the vertex set V:

- $E_{\text{intra}} = \{(u, v) \in E : \exists i \text{ such that } u, v \in C_i\}$ (intra-community edges)
- $E_{\text{inter}} = \{(u, v) \in E : u \in C_i, v \in C_j, i \neq j\}$ (inter-community edges)

Let $n_1 = |E_{\text{intra}}|$, $n_2 = |E_{\text{inter}}|$, and $m = n_1 + n_2$.

Definition 3 (Mean DSpar Scores). The mean DSpar scores for each edge type are:

$$\mu_{\text{intra}} = \frac{1}{n_1} \sum_{e \in E_{\text{intra}}} s(e), \qquad \mu_{\text{inter}} = \frac{1}{n_2} \sum_{e \in E_{\text{inter}}} s(e) \tag{12}$$

The global mean is $\bar{s} = \frac{1}{m} \sum_{e \in E} s(e) = \frac{n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}}{m}$.

Definition 4 (DSpar Separation). A partition $\mathcal{C}$ exhibits DSpar separation with gap $\delta > 0$ if:

$$\mu_{\text{intra}} - \mu_{\text{inter}} = \delta > 0 \tag{13}$$

Definition 5 (Calibrated DSpar Sparsification). DSpar sparsification with retention parameter $\alpha \in (0, 1]$ retains each edge e independently with probability

$$p(e) = \min\bigl(1,\ \lambda_\alpha\, s(e)\bigr), \tag{14}$$

where λ_α solves $\sum_{e \in E} \min(1, \lambda_\alpha s(e)) = \alpha m$, so that the expected number of retained edges equals αm exactly. When $\lambda_\alpha s_{\max} \leq 1$ (no probability clips), this reduces to $p(e) = \alpha\, s(e)/\bar{s}$. For $\alpha \in (0, 1)$ the map $\lambda \mapsto \sum_e \min(1, \lambda s(e))$ is continuous and strictly increasing below saturation, so λ_α exists and is unique; $\alpha = 1$ denotes the identity (no sparsification).

Remark 1 (Clipping). Our formal results assume the unclipped regime $\lambda_\alpha s_{\max} \leq 1$, in which retention probabilities are exactly proportional to scores. The condition is stringent on real networks: it requires $\alpha \leq \bar{s}/s_{\max}$, roughly 0.05 on email-Eu-core; at the retention levels used in practice, and in our experiments, a majority of low-degree edges can clip at $p(e) = 1$ (about 65% on email-Eu-core at calibrated $\alpha = 0.9$). Because $\min(1, \lambda s)$ is concave in s, $\delta > 0$ alone then no longer implies even the direction of preferential removal; a sufficient additional hypothesis is first-order stochastic dominance of the intra-community score distribution over the inter-community one, under which any nondecreasing retention rule preserves the direction. The formal results below therefore describe the score-proportional regime. Section 4 audits this gap, together with the larger gap between nominal and realized retention in previously published sampler variants; the direction of preferential removal is confirmed empirically on all networks whose ΔF_{obs} is distinguishable from zero (Section 5.1).

Definition 6 (Edge Preservation Ratio). After sparsification, let $E' \subseteq E$ denote the retained edges. The edge preservation ratio is:

$$\text{Ratio} = \frac{|E' \cap E_{\text{inter}}|/n_2}{|E' \cap E_{\text{intra}}|/n_1} \tag{15}$$

A ratio less than 1 indicates preferential removal of inter-community edges; the ratio is defined whenever some intra-community edge is retained.

Assumption 1 (Non-degeneracy). We assume throughout that:

1) $n_1 \geq 1$ and $n_2 \geq 1$ (both edge types exist)
2) $\mu_{\text{intra}} > 0$ and $\mu_{\text{inter}} > 0$ (positive mean scores)

These conditions hold whenever the partition places at least one edge inside a community and at least one edge between communities; positivity of the mean scores is automatic since $s(e) > 0$ for every edge.

Theorem 1 (DSpar Separation Implies Preferential Removal). Let G be a graph with partition $\mathcal{C}$ exhibiting DSpar separation with gap $\delta > 0$. Under DSpar sparsification in the unclipped regime ($\lambda_\alpha s_{\max} \leq 1$, Remark 1):

(a) The ratio of expected preservation rates satisfies:

$$\frac{\mathbb{E}[|E' \cap E_{\text{inter}}|]/n_2}{\mathbb{E}[|E' \cap E_{\text{intra}}|]/n_1} = \frac{\mu_{\text{inter}}}{\mu_{\text{intra}}} < 1 \tag{16}$$

(b) Asymptotic Convergence: Consider a sequence of graphs $\{G_n\}_{n=1}^{\infty}$ with partitions $\{\mathcal{C}_n\}$ satisfying:

(i) $n_1^{(n)}, n_2^{(n)} \to \infty$ as $n \to \infty$
(ii) The mean scores converge: $\mu_{\text{intra}}^{(n)} \to \mu_{\text{intra}}^*$ and $\mu_{\text{inter}}^{(n)} \to \mu_{\text{inter}}^*$ for constants $\mu_{\text{intra}}^* > \mu_{\text{inter}}^* > 0$ (which supplies uniform bounds $\mu_{\min}, \mu_{\max}$ and a maintained separation $\delta_0 > 0$)
(iii) The unclipped condition holds along the sequence: $\lambda_\alpha^{(n)} s_{\max}^{(n)} \leq 1$ for all n

Then the edge preservation ratio converges in probability:

$$\text{Ratio}_n \xrightarrow{P} \frac{\mu_{\text{inter}}^*}{\mu_{\text{intra}}^*} < 1 \tag{17}$$

(c) Concentration Bound: For any $\epsilon > 0$:

$$\mathbb{P}\left[\left|\frac{|E' \cap E_{\text{inter}}|}{n_2} - \frac{\alpha \mu_{\text{inter}}}{\bar{s}}\right| > \epsilon\right] \leq \frac{\alpha \mu_{\text{inter}}}{n_2 \bar{s} \epsilon^2} \tag{18}$$

and similarly for the intra-community preservation rate.

Proof. Part (a): Let $X_{\text{inter}} = |E' \cap E_{\text{inter}}|$ denote the number of preserved inter-community edges. Since edges are retained independently with probability $p(e) = \alpha s(e)/\bar{s}$:

$$\begin{aligned} \mathbb{E}[X_{\text{inter}}] &= \sum_{e \in E_{\text{inter}}} p(e) \\ &= \sum_{e \in E_{\text{inter}}} \frac{\alpha\, s(e)}{\bar{s}} \\ &= \frac{\alpha}{\bar{s}} \sum_{e \in E_{\text{inter}}} s(e) \\ &= \frac{\alpha \cdot n_2 \cdot \mu_{\text{inter}}}{\bar{s}} \end{aligned} \tag{19}$$

The expected inter-community preservation rate is:

$$\frac{\mathbb{E}[X_{\text{inter}}]}{n_2} = \frac{\alpha \cdot \mu_{\text{inter}}}{\bar{s}} \tag{20}$$

By identical reasoning:

$$\frac{\mathbb{E}[X_{\text{intra}}]}{n_1} = \frac{\alpha \cdot \mu_{\text{intra}}}{\bar{s}} \tag{21}$$

Taking the ratio:

$$\frac{\mathbb{E}[X_{\text{inter}}]/n_2}{\mathbb{E}[X_{\text{intra}}]/n_1} = \frac{\alpha\mu_{\text{inter}}/\bar{s}}{\alpha\mu_{\text{intra}}/\bar{s}} = \frac{\mu_{\text{inter}}}{\mu_{\text{intra}}} < 1 \tag{22}$$

where the inequality follows from DSpar separation ($\mu_{\text{intra}} > \mu_{\text{inter}}$).

Part (b): Define empirical preservation rates:

$$\hat{p}_{\text{inter}} = \frac{X_{\text{inter}}}{n_2}, \qquad \hat{p}_{\text{intra}} = \frac{X_{\text{intra}}}{n_1} \tag{23}$$

From the concentration bound in part (c), for any $\epsilon > 0$:

$$\mathbb{P}\left[|\hat{p}_{\text{inter}} - \mathbb{E}[\hat{p}_{\text{inter}}]| > \epsilon\right] \leq \frac{\alpha\mu_{\text{inter}}}{n_2\bar{s}\epsilon^2} \to 0 \quad \text{as } n_2 \to \infty \tag{24}$$

Hence $\hat{p}_{\text{inter}} = \frac{\alpha\mu_{\text{inter}}^{(n)}}{\bar{s}^{(n)}}\big(1 + o_P(1)\big)$ and $\hat{p}_{\text{intra}} = \frac{\alpha\mu_{\text{intra}}^{(n)}}{\bar{s}^{(n)}}\big(1 + o_P(1)\big)$; note that $\bar{s}^{(n)}$, a convex combination of $\mu_{\text{intra}}^{(n)}$ and $\mu_{\text{inter}}^{(n)}$, is bounded in $[\mu_{\min}, \mu_{\max}]$, and it cancels in the ratio.

To apply the Continuous Mapping Theorem to the ratio, we must verify that $\hat{p}_{\text{intra}}$ is bounded away from zero with high probability. Since $\mu_{\text{intra}} \geq \mu_{\min} > 0$ and $\bar{s} \leq 2$ (as $s(e) \leq 2$ for any edge), we have $\mathbb{E}[\hat{p}_{\text{intra}}] \geq \alpha\mu_{\min}/2 > 0$. Taking $\epsilon = \alpha\mu_{\min}/(4)$, the concentration bound gives:

$$\mathbb{P}\left[\hat{p}_{\text{intra}} < \frac{\alpha\mu_{\min}}{4}\right] \leq \mathbb{P}\left[|\hat{p}_{\text{intra}} - \mathbb{E}[\hat{p}_{\text{intra}}]| > \frac{\alpha\mu_{\min}}{4}\right] \to 0 \tag{25}$$

Thus $\hat{p}_{\text{intra}}$ is bounded away from zero with probability approaching 1. The Continuous Mapping Theorem (applied to $g(x, y) = x/y$, continuous on $\{(x, y) : y > 0\}$) yields:

$$\text{Ratio}_n = \frac{\hat{p}_{\text{inter}}}{\hat{p}_{\text{intra}}} \xrightarrow{P} \frac{\mu^*_{\text{inter}}}{\mu^*_{\text{intra}}} < 1 \tag{26}$$

Part (c): Let $Y = |E' \cap E_{\text{inter}}| = \sum_{e \in E_{\text{inter}}} X_e$ where $X_e = \mathbf{1}[e \in E']$.

Since the X_e are independent Bernoulli random variables:

$$\text{Var}(Y) = \sum_{e \in E_{\text{inter}}} p(e)(1 - p(e)) \leq \sum_{e \in E_{\text{inter}}} p(e) = \frac{\alpha n_2 \mu_{\text{inter}}}{\bar{s}} \tag{27}$$

By Chebyshev's inequality:

$$\mathbb{P}[|Y - \mathbb{E}[Y]| > n_2\epsilon] \leq \frac{\text{Var}(Y)}{n_2^2\epsilon^2} \leq \frac{\alpha\mu_{\text{inter}}}{n_2\bar{s}\epsilon^2} \tag{28}$$

□

Corollary 1 (Quantifying Preferential Removal). The ratio of expected preservation rates in Theorem 1(a) equals

$$1 - \frac{\delta}{\mu_{\text{intra}}}, \tag{29}$$

where $\delta = \mu_{\text{intra}} - \mu_{\text{inter}}$ is the DSpar separation gap; under the hypotheses of Theorem 1(b), the empirical ratio converges in probability to the corresponding limiting value. Larger gaps yield smaller ratios and stronger preferential removal.

Proof. From Theorem 1(a), the ratio is $\mu_{\text{inter}}/\mu_{\text{intra}}$. Substituting $\mu_{\text{inter}} = \mu_{\text{intra}} - \delta$:

$$\frac{\mu_{\text{inter}}}{\mu_{\text{intra}}} = \frac{\mu_{\text{intra}} - \delta}{\mu_{\text{intra}}} = 1 - \frac{\delta}{\mu_{\text{intra}}} \tag{30}$$

□

Remark 2 (Relationship to Hub-Bridging). A common structural property in real networks is hub-bridging: inter-community edges tend to connect higher-degree nodes. Formally, $\mathbb{E}[d_u d_v \mid e \in E_{\text{inter}}] > \mathbb{E}[d_u d_v \mid e \in E_{\text{intra}}]$. Hub-bridging and DSpar separation are related but distinct: hub-bridging concerns degree products, while DSpar separation concerns inverse-degree sums. Empirically, the two conditions are often correlated in the networks we study: networks in which high-degree nodes disproportionately bridge communities tend to exhibit DSpar separation, in the sense that inter-community edges receive lower $1/d_u + 1/d_v$ scores on average. Empirically the two conditions frequently co-occur but can dissociate: Section 5 reports real networks with hub-bridging ratio below one yet $\delta > 0$, and configuration-model graphs exhibiting both conditions despite having no community structure.

Definition 7 (Weighted Modularity). For a graph with edge weights $w_e \geq 0$, the modularity with respect to partition $\mathcal{C}$ is:

$$Q^{(w)} = \frac{W_{\text{intra}}}{W} - \sum_{c \in \mathcal{C}} \left(\frac{D_c^{(w)}}{2W}\right)^2 \tag{31}$$

where:

- $W = \sum_{e \in E} w_e$ is the total edge weight
- $W_{\text{intra}} = \sum_{e \in E_{\text{intra}}} w_e$ is the intra-community edge weight
- $D_c^{(w)} = \sum_{i \in c} d_i^{(w)}$ is the total weighted degree of community c
- $d_i^{(w)} = \sum_{j:(i,j) \in E} w_{(i,j)}$ is the weighted degree of node i, where $w_{(i,j)}$ denotes the weight of edge (i, j)

We write $Q^{(w)} = F^{(w)} - G^{(w)}$ where $F^{(w)} = W_{\text{intra}}/W$ and $G^{(w)} = \sum_c (D_c^{(w)}/2W)^2$.

Definition 8 (DSpar Weighting). The DSpar weighting assigns weight $w_e = s(e)/\bar{s}$ to each edge, where $s(e)$ is the DSpar score and $\bar{s}$ is the mean. This normalizes the total weight to equal the edge count: $W = m$.

Definition 9 (Balanced Communities). A partition $\mathcal{C}$ has balanced communities if all communities have equal total degree:

$$D_c = \frac{2m}{k} \quad \text{for all } c \in \mathcal{C} \tag{32}$$

where $k = |\mathcal{C}|$ is the number of communities.

Theorem 2 (Modularity Improvement under DSpar). Let G be a graph with partition $\mathcal{C}$ exhibiting DSpar separation. Let $Q^{(1)}$ denote modularity under unit weights and $Q^{(s)}$ denote modularity under DSpar weighting.

(a) Intra-Community Fraction Improvement: The weighted intra-community fraction strictly increases:

$$F^{(s)} - F^{(1)} = \frac{n_1 n_2(\mu_{\text{intra}} - \mu_{\text{inter}})}{m(n_1\mu_{\text{intra}} + n_2\mu_{\text{inter}})} > 0 \tag{33}$$

(b) Modularity Decomposition: The modularity change decomposes as:

$$Q^{(s)} - Q^{(1)} = \underbrace{(F^{(s)} - F^{(1)})}_{>0 \text{ by part (a)}} - \underbrace{(G^{(s)} - G^{(1)})}_{\text{sign depends on network}} \tag{34}$$

(c) Sufficient Condition: $Q^{(s)} > Q^{(1)}$ if and only if $G^{(s)} - G^{(1)} < F^{(s)} - F^{(1)}$. In particular, $Q^{(s)} > Q^{(1)}$ whenever $G^{(s)} \leq G^{(1)}$.

Proof. Part (a): Under unit weights, the intra-community fraction is:

$$F^{(1)} = \frac{n_1}{m} = \frac{n_1}{n_1 + n_2} \tag{35}$$

Under DSpar weighting with $w_e = s(e)/\bar{s}$, the normalization $\bar{s}$ cancels between numerator and denominator:

$$F^{(s)} = \frac{\sum_{e \in E_{\text{intra}}} s(e)}{\sum_{e \in E} s(e)} = \frac{n_1 \mu_{\text{intra}}}{n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}} \tag{36}$$

We show $F^{(s)} > F^{(1)}$. This inequality is equivalent to:

$$\frac{n_1 \mu_{\text{intra}}}{n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}} > \frac{n_1}{n_1 + n_2} \tag{37}$$

Cross-multiplying (all terms positive):

$$n_1 \mu_{\text{intra}}(n_1 + n_2) > n_1(n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}) \tag{38}$$

$$n_1^2 \mu_{\text{intra}} + n_1 n_2 \mu_{\text{intra}} > n_1^2 \mu_{\text{intra}} + n_1 n_2 \mu_{\text{inter}} \tag{39}$$

$$n_1 n_2 \mu_{\text{intra}} > n_1 n_2 \mu_{\text{inter}} \tag{40}$$

$$\mu_{\text{intra}} > \mu_{\text{inter}} \tag{41}$$

which holds by DSpar separation.

The explicit formula follows by computing $F^{(s)} - F^{(1)}$ directly:

$$F^{(s)} - F^{(1)} = \frac{n_1 \mu_{\text{intra}}}{n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}}} - \frac{n_1}{n_1 + n_2} \tag{42}$$

$$= n_1 \cdot \frac{\mu_{\text{intra}}(n_1 + n_2) - (n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}})}{(n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}})(n_1 + n_2)} \tag{43}$$

$$= \frac{n_1 n_2 (\mu_{\text{intra}} - \mu_{\text{inter}})}{m(n_1 \mu_{\text{intra}} + n_2 \mu_{\text{inter}})} \tag{44}$$

Parts (b) and (c): Follow directly from the definition $Q = F - G$. □

Lemma 1 (Characterization of G Change). Under DSpar weighting (which by construction preserves total weight: $W^{(s)} = W^{(1)} = m$):

$$G^{(s)} - G^{(1)} = \frac{1}{4m^2} \left[\sum_c (D_c^{(s)})^2 - \sum_c (D_c^{(1)})^2 \right] \tag{45}$$

Thus $G^{(s)} \leq G^{(1)}$ if and only if $\sum_c (D_c^{(s)})^2 \leq \sum_c (D_c^{(1)})^2$.

Proof. By definition:

$$G^{(w)} = \sum_c \left(\frac{D_c^{(w)}}{2W} \right)^2 = \frac{1}{4W^2} \sum_c \left(D_c^{(w)}\right)^2 \tag{46}$$

Since $W^{(s)} = W^{(1)} = m$:

$$G^{(s)} - G^{(1)} = \frac{1}{4m^2} \sum_c \left(D_c^{(s)}\right)^2 - \frac{1}{4m^2} \sum_c \left(D_c^{(1)}\right)^2 = \frac{1}{4m^2} \left[\sum_c \left(D_c^{(s)}\right)^2 - \sum_c \left(D_c^{(1)}\right)^2 \right] \tag{47}$$

□

Remark 3 (When $G^{(s)} \leq G^{(1)}$). DSpar weighting assigns higher weights to edges connecting low-degree nodes. In networks with hub-bridging structure, high-degree nodes have edges with low DSpar scores. This tends to compress the weighted degree distribution, reducing the variance of $D_c^{(s)}$ across communities, which implies $G^{(s)} \leq G^{(1)}$. Our experiments confirm this condition holds across all tested networks.

3.3 Sufficient Conditions for $G^{(s)} \leq G^{(1)}$

By Theorem 2(c), modularity improves whenever $G^{(s)} \leq G^{(1)}$. By Lemma 1, this is equivalent to $\sum_c (D_c^{(s)})^2 \leq \sum_c (D_c^{(1)})^2$. We provide verifiable conditions:

Proposition 1 (Verifiable Conditions for $G^{(s)} \leq G^{(1)}$). (A) Direct Verification: Compute $D_c^{(s)} = \sum_{i \in c} d_i^{(s)}$ for each community, where:

$$d_i^{(s)} = \sum_{j \in N(i)} \frac{s(i,j)}{\bar{s}} = \frac{1}{\bar{s}} \left(1 + \sum_{j \in N(i)} \frac{1}{d_j} \right) \tag{48}$$

Then $G^{(s)} \leq G^{(1)}$ if and only if:

$$\sum_{c \in \mathcal{C}} (D_c^{(s)})^2 \leq \sum_{c \in \mathcal{C}} (D_c^{(1)})^2 \tag{49}$$

This condition can be checked numerically for any given graph and partition.

(B) Variance Form: Equivalently, since the total degree is fixed under both weightings, the sum of squares does not increase if and only if the variance of community degree sums does not increase:

$$\text{Var}_c(D_c^{(s)}) \leq \text{Var}_c(D_c^{(1)}) \tag{50}$$

where variance is taken over communities. Since $\sum_c D_c^{(s)} = \sum_c D_c^{(1)} = 2m$ (total degree is preserved), reduced variance and reduced sum of squares are equivalent.

(C) Perfectly Balanced Communities: If $D_c^{(1)} = 2m/k$ for all communities c (perfectly balanced), then $G^{(1)} = 1/k$. If additionally $D_c^{(s)} = 2m/k$ for all c, then $G^{(s)} = G^{(1)} = 1/k$. Note: This second condition rarely holds exactly, but approximate balance ($D_c^{(s)} \approx 2m/k$) yields $G^{(s)} \approx G^{(1)}$.

Proof. (A) follows directly from Lemma 1.

(B) For any set of values $\{x_c\}$ with fixed sum $S = \sum_c x_c$, we have $\sum_c x_c^2 = k \cdot \text{Var}_c(x_c) + S^2/k$. Since $\sum_c D_c^{(s)} = \sum_c D_c^{(1)} = 2m$, variance reduction and sum-of-squares reduction are equivalent.

(C) If $D_c = 2m/k$ for all c, then $\sum_c D_c^2 = k \cdot (2m/k)^2 = 4m^2/k$, so $G = 1/k$. □

Remark 4 (Why $G^{(s)} \leq G^{(1)}$ Often Holds). The weighted degree formula shows that $d_i^{(s)}$ depends on the degrees of i's neighbors. High-degree hubs connected to other hubs have $d_i^{(s)} < d_i^{(1)}$ (their edge weights are reduced). In networks where hubs are concentrated in a few communities, this compression reduces the variance of $D_c^{(s)}$, satisfying condition (B). Our experiments confirm $G^{(s)} \leq G^{(1)}$ holds across all tested networks.

Corollary 2 (Quantitative Improvement Bound). Assume communities are balanced under both weightings, i.e., $D_c^{(1)} =$

$D_c^{(s)} = 2m/k$ for all c. Then the modularity improvement satisfies:

$$Q^{(s)} - Q^{(1)} \geq \frac{n_1 n_2 \delta}{m(n_1\mu_{\text{intra}} + n_2\mu_{\text{inter}})} \tag{51}$$

where $\delta = \mu_{\text{intra}} - \mu_{\text{inter}}$ is the DSpar separation gap.

Proof. Under the stated assumption, $G^{(s)} = G^{(1)} = 1/k$. Therefore:

$$Q^{(s)} - Q^{(1)} = (F^{(s)} - G^{(s)}) - (F^{(1)} - G^{(1)}) = F^{(s)} - F^{(1)} \tag{52}$$

The result follows from Theorem 2(a). □

The weighted-balance assumption is restrictive and rarely holds exactly (Proposition 1(C)); we use this corollary for intuition rather than practice.

While Theorems 1 and 2 analyze DSpar weighting, our experiments use DSpar sparsification (edge removal). The following theorem directly addresses sparsification:

Theorem 3 (Sparsification Improves Intra-Community Fraction (Asymptotically)). Let G be a graph with partition $\mathcal{C}$ exhibiting DSpar separation, and let G' denote the DSpar-sparsified graph (unclipped regime, Remark 1).

(a) High-Probability Approximation: Along any sequence of graphs satisfying conditions (i)–(ii) of Theorem 1(b), for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|F' - \frac{n_1\mu_{\text{intra}}}{n_1\mu_{\text{intra}} + n_2\mu_{\text{inter}}}\right| > \varepsilon\right) \longrightarrow 0 \quad \text{as } n_1, n_2 \to \infty. \tag{53}$$

In particular, $F' = F^{(s)} + o_p(1)$.

Here $o_p(1)$ denotes a random term converging to zero in probability as the graph size increases; consequently, F' concentrates around the deterministic limit $F^{(s)}$.

(b) Asymptotic Improvement: Consider a sequence of graphs $\{G_n\}$ satisfying the conditions of Theorem 1(b), and assume additionally that $n_2^{(n)}/n_1^{(n)} \to \rho > 0$. Then

$$F'_n \xrightarrow{P} \frac{\mu^*_{\text{intra}}}{\mu^*_{\text{intra}} + \rho\mu^*_{\text{inter}}} > \frac{1}{1+\rho} = \lim_{n\to\infty} F_n^{(1)}, \tag{54}$$

where the inequality follows from preserved DSpar separation ($\mu^*_{\text{intra}} > \mu^*_{\text{inter}}$).

Proof. Let $X_{\text{intra}} = |E' \cap E_{\text{intra}}|$ and $X_{\text{inter}} = |E' \cap E_{\text{inter}}|$. By Theorem 1(a),

$$\mathbb{E}[X_{\text{intra}}] = \frac{\alpha n_1 \mu_{\text{intra}}}{\bar{s}}, \qquad \mathbb{E}[X_{\text{inter}}] = \frac{\alpha n_2 \mu_{\text{inter}}}{\bar{s}}. \tag{55}$$

From the concentration bound in Theorem 1(c), both X_{intra}/n_1 and X_{inter}/n_2 converge in probability to their respective expectations. Writing

$$F' = \frac{X_{\text{intra}}/n_1}{X_{\text{intra}}/n_1 + (n_2/n_1)(X_{\text{inter}}/n_2)}, \tag{56}$$

and applying Slutsky's theorem together with the Continuous Mapping Theorem yields part (a).

For part (b), the assumed convergence $n_2/n_1 \to \rho$ gives

$$F'_n \xrightarrow{P} \frac{\mu^*_{\text{intra}}}{\mu^*_{\text{intra}} + \rho\mu^*_{\text{inter}}}. \tag{57}$$

Since $F_n^{(1)} = 1/(1 + n_2/n_1) \to 1/(1+\rho)$ and $\mu^*_{\text{intra}} > \mu^*_{\text{inter}}$, the claimed inequality follows. □

Remark 5 (Connection Between Weighting and Sparsification). Theorem 3 shows that DSpar sparsification produces, with high probability in large graphs, the same intra-community fraction as DSpar weighting. This establishes a formal link between the deterministic weighting analysis (Theorem 2) and the randomized sparsification procedure used in our experiments.

3.4 Connection to Spectral Theory

Newman [32] established a close relationship between modularity maximization and spectral properties of degree-normalized graph operators. We summarize the aspects most relevant to our analysis.

Proposition 2 (Spectral Relaxation of Modularity [14], [32]). In the bipartition setting, modularity maximization admits a standard spectral relaxation in which the discrete constraint $s_i \in \{-1, +1\}$ is relaxed to continuous values subject to degree-weighted normalization and orthogonality constraints (e.g., $k^\top s = 0$). The relaxed problem reduces to a leading eigenvector computation of a modularity-based operator, which can be expressed as a generalized eigenvalue problem of the form

$$Bs = \lambda Ds, \tag{58}$$

or, equivalently under the orthogonality constraint, as

$$As = \lambda Ds. \tag{59}$$

The optimal relaxed solution is proportional to the corresponding leading eigenvector, and a discrete partition is obtained by rounding (e.g., by taking signs). The achieved relaxed modularity is proportional to the associated eigenvalue, with the constant of proportionality depending on the specific normalization.

This result provides useful spectral intuition for our approach:

1) Spectral sparsification [3] preserves quadratic forms and, consequently, the eigenvalues of degree-normalized graph operators up to $(1 \pm \varepsilon)$ factors by sampling edges proportional to effective resistance.
2) DSpar is a degree-based proxy motivated by classical connections between effective resistance, random walks, and graph connectivity [3], [22]. While DSpar does not provide worst-case spectral approximation guarantees, it preferentially downweights edges incident to high-degree nodes, similar to resistance-based sampling in heterogeneous networks.
3) Spectral interpretation of communities. For degree-normalized operators such as the normalized adjacency matrix $\mathcal{A} = D^{-1/2}AD^{-1/2}$, community structure corresponds to slow mixing of random walks between groups. In this setting, a second eigenvalue λ_2 close to 1 indicates strong separation between communities.

Taken together, these observations suggest that sparsification schemes which preferentially remove inter-community edges while preserving intra-community connectivity tend to preserve or even accentuate the spectral signatures associated with community structure. In particular, by preferentially removing inter-community edges (Theorem 1), DSpar makes the graph closer to block-diagonal form. In the idealized block-diagonal limit of completely disconnected communities, the normalized adjacency matrix has multiple eigenvalues equal to 1, corresponding to perfectly separated components.

Remark 6 (Random-Walk Interpretation). For the normalized adjacency matrix $\mathcal{A}$, eigenvalues near 1 correspond to slowly mixing random walks. Reducing inter-community connectivity increases the persistence of random walks within communities, providing a spectral interpretation of improved community separation. This interpretation is heuristic but aligns with both classical spectral clustering theory and our empirical observations.

The key insight is that DSpar separation, where intra-community edges receive systematically higher DSpar scores than inter-community edges, is sufficient to guarantee preferential removal of inter-community edges (Theorem 1) and an increase in the intra-community edge fraction, both under weighting (Theorem 2(a)) and sparsification (Theorem 3). For full modularity improvement under weighting, we additionally require $G^{(s)} \leq G^{(1)}$, which holds empirically across all tested networks. The spectral perspective provides additional intuition but is not required for the formal results.

4 Sampler Audit: Nominal versus Realized Retention

Before presenting experiments, we audit what DSpar-style samplers actually retain, because the retention parameter α has been used with three mutually inconsistent meanings in the literature, including in an earlier version of this work.

4.0.0.1 With-replacement sampling collapses retention.: The originally published DSpar procedure [6] draws $q = \lceil \alpha m \rceil$ edge samples with replacement from the distribution proportional to $s(e)$, reweights, and keeps the union of sampled edges. Because high-score edges are drawn repeatedly, the number of distinct retained edges is far below αm: across our 17-network suite, nominal $\alpha = 0.8$ realizes only 0.32–0.55 of the edges (Table 1). Results reported at "$\alpha = 0.8$" under this sampler are, in fact, roughly half-density experiments.

4.0.0.2 Probability clipping causes saturation.: A natural no-replacement variant retains edge e independently with probability proportional to its score, clipped at one. Because a substantial fraction of low-degree edges clips (12–36% in our measurements), the expected retention falls strictly below αm at every setting and saturates: on email-Eu-core this sampler cannot retain more than 66% of edges even at $\alpha = 1.0$. True mild sparsification (e.g., retaining 90% of edges) is unreachable under either published variant.

4.0.0.3 Calibration closes the gap.: Definition 5 instead solves for the proportionality constant λ_α by bisection so that expected retention equals αm exactly. Measured realized retention under the calibrated sampler matches the nominal value to within ± 0.001 on the mid-sized networks at every tested setting ($\alpha \in \{0.7, 0.8, 0.9, 0.95\}$), and to within ± 0.013 on the smallest graphs. We verified our implementations of the with-replacement variant to be bit-identical to the repository implementation that produced the earlier experiments before use.

4.0.0.4 Consequences.: Three practices follow, which we adopt throughout. First, realized retention m'/m is reported alongside nominal α in every experiment. Second, comparisons across sampler variants are made at matched realized retention, not matched nominal α. Third, we reserve $\alpha = 1.0$ strictly as the no-sparsification baseline; note that under the with-replacement variant even $\alpha = 1.0$ removes roughly half the edges, so pipelines that treat it as a baseline must bypass the sampler explicitly.

TABLE 1
Sampler audit: nominal retention parameter α versus realized distinct-edge retention m'/m (min–max of per-network means; "#" = pooled runs). The with-replacement sampler collapses retention to roughly $\alpha/2$ even at $\alpha = 1.0$; the clipped no-replacement variant saturates below 0.7; only the calibrated sampler (Definition 5) realizes its nominal rate. The calibrated $\alpha = 1.0$ cell is by construction the no-sparsification baseline.

Sampler	nominal α	realized m'/m	#
with replacement	0.8	0.317–0.549	54
	0.9	0.334–0.554	15
	1.0	0.350–0.589	15
no replacement (clipped)	0.8	0.400–0.619	6
	0.9	0.432–0.660	6
	1.0	0.662	1
calibrated (Def. 5)	0.8	0.799–0.800	6
	0.9	0.887–0.901	26
	1.0	–	–

5 The Mechanism and Its Null Models

This section verifies the fixed-partition theory quantitatively (§5.1), then asks what the verified signals mean, using degree-preserving configuration models (§5.2) and partition-provenance controls (§5.3).

5.1 Fixed-Partition Decomposition on Real Networks

5.1.0.1 Setup.: For each network we obtain a partition with the Leiden algorithm [11] on the original graph and hold it fixed; all quantities below are changes induced by sparsification alone, with no re-optimization. We apply the with-replacement DSpar sampler at nominal $\alpha = 0.8$ (realized retention 0.33–0.48 on these eleven networks; Section 4), 10 independent runs, and evaluate the fixed partition on the sparsified graph. Writing $Q = F - G$ for modularity, with F the intra-community edge fraction and G the null-model penalty, we verify the identity $\Delta Q_{\text{fixed}} = \Delta F_{\text{obs}} - \Delta G_{\text{obs}}$ to machine precision on every run. All experiments in this paper use the largest connected component of each simple undirected graph. The Artifact I sweep (Section 6.1) and the boundary study (Section 7) likewise use overlapping but not identical 15-network subsets of the 17-network suite; each table lists its own membership.

5.1.0.2 What the measurements show.: Table 2 shows positive ΔQ_{fixed} on all eleven networks; Table 3 reports the decomposition into ΔF_{obs} and $-\Delta G_{\text{obs}}$. The identity itself is algebraic; the content lies in the sizes and signs of the two terms. Two regimes appear, consistent with Theorem 2(b): in networks such as cit-HepTh and ca-CondMat the intra-community fraction rises ($\Delta F_{\text{obs}} > 0$), consistent with preferential inter-community removal under $\delta > 0$; in facebook-combined ($\delta \approx 0$) and ca-HepPh the change is carried almost entirely by the null-model term, with ΔF_{obs} statistically

TABLE 2
Fixed-partition modularity changes under with-replacement DSpar at nominal $\alpha = 0.8$, computed on the largest connected component of each simple undirected graph (realized retention 0.33–0.48). Both ΔQ columns are evaluated on the sparsified graph; Section 6 shows why such cross-graph quantities must not be read as detection improvements.

Dataset	m	m'/m	Q_{orig}	ΔQ_{fixed}	$\Delta Q^{(\text{sp})}_{\text{Leiden}}$
ca-AstroPh [35]	196,972	0.4166	0.6343	0.0399 ± 0.0007	0.0529 ± 0.0012
ca-CondMat [35]	91,286	0.4715	0.7312	0.0614 ± 0.0015	0.0999 ± 0.0014
ca-GrQc [35]	13,422	0.4545	0.8514	0.0187 ± 0.0022	0.0540 ± 0.0024
ca-HepPh [35]	117,619	0.3399	0.6512	0.0848 ± 0.0008	0.1092 ± 0.0015
ca-HepTh [35]	24,806	0.4766	0.7619	0.0542 ± 0.0015	0.1134 ± 0.0019
cit-HepPh [35]	420,784	0.4593	0.7314	0.0120 ± 0.0004	0.0188 ± 0.0022
cit-HepTh [35]	352,021	0.4332	0.6567	0.0467 ± 0.0007	0.0582 ± 0.0021
email-Enron [36]	180,811	0.3886	0.6124	0.1429 ± 0.0012	0.1798 ± 0.0020
facebook-combined [37]	88,234	0.4360	0.8355	0.0253 ± 0.0005	0.0277 ± 0.0010
wiki-Vote [38]	100,736	0.3328	0.4240	0.0734 ± 0.0012	0.0827 ± 0.0077
email-Eu-core [39]	16,064	0.4429	0.4159	0.0819 ± 0.0025	0.0919 ± 0.0034

TABLE 3
Modularity decomposition verification at nominal $\alpha = 0.8$ (largest connected component of each graph). For all datasets and runs, the identity $\Delta Q_{\text{fixed}} = \Delta F_{\text{obs}} - \Delta G_{\text{obs}}$ holds up to machine precision (maximum absolute reconstruction error $\leq 10^{-14}$).

Dataset	ΔQ_{fixed}	ΔF_{obs}	$-\Delta G_{\text{obs}}$
ca-AstroPh	0.0399 ± 0.0007	0.0337 ± 0.0007	0.0062 ± 0.0001
ca-CondMat	0.0614 ± 0.0015	0.0584 ± 0.0015	0.0030 ± 0.0000
ca-GrQc	0.0187 ± 0.0022	0.0083 ± 0.0022	0.0105 ± 0.0001
ca-HepPh	0.0848 ± 0.0008	-0.0111 ± 0.0009	0.0959 ± 0.0002
ca-HepTh	0.0542 ± 0.0015	0.0499 ± 0.0015	0.0043 ± 0.0002
cit-HepPh	0.0120 ± 0.0004	0.0084 ± 0.0004	0.0037 ± 0.0001
cit-HepTh	0.0467 ± 0.0007	0.0350 ± 0.0007	0.0117 ± 0.0001
email-Enron	0.1429 ± 0.0012	0.1086 ± 0.0011	0.0343 ± 0.0002
facebook-combined	0.0253 ± 0.0005	-0.0006 ± 0.0006	0.0259 ± 0.0002
wiki-Vote	0.0734 ± 0.0012	0.0726 ± 0.0014	0.0008 ± 0.0002
email-Eu-core	0.0819 ± 0.0025	0.0734 ± 0.0027	0.0085 ± 0.0009

indistinguishable from zero. Figures 1 and 2 illustrate both quantities across retention levels; all effects vanish smoothly as $\alpha \to 1$. Because these runs use the with-replacement sampler, the Bernoulli-model ratio of Corollary 1 is not expected to hold exactly, and it does not: measured preservation ratios are closer to one than predicted on all eleven networks, by 0.01 to 0.29, which is the quantitative signature of that sampler's concave retention profile (Section 4). The direction of preferential removal agrees with the prediction on nine of eleven networks; the two exceptions are exactly the two networks whose ΔF_{obs} is indistinguishable from zero.

5.1.0.3 What is not shown.: These measurements verify the mechanism of Section 3: degree-biased sampling shifts edge composition and null-model mass exactly as predicted. They do not show that community detection improved. Both ΔQ columns compare modularity across different graphs (the fixed partition scored on G versus on the sparsified G'), and Section 6 demonstrates that this comparison is invalid as a quality metric. Presented without controls, such measurements read as evidence of improvement; the controls below explain why that reading fails.

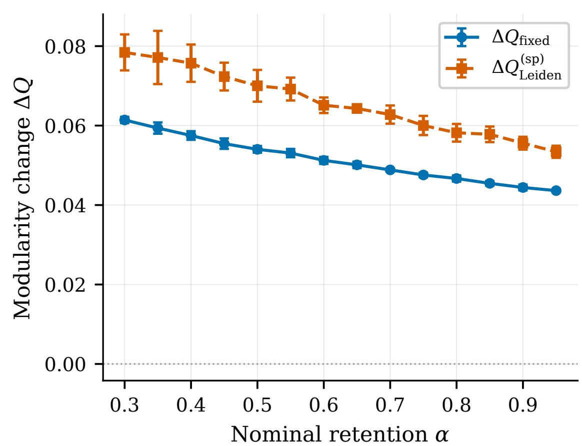

Fig. 1. Fixed-partition and re-optimized modularity changes under DSpar sparsification on cit-HepTh, both evaluated on the sparsified graph.

5.2 Configuration-Model Controls

If $\delta > 0$ and $\Delta Q_{\text{fixed}} > 0$ reflected community structure, they should weaken or vanish on graphs whose communities have been destroyed. We test this directly: each network is degree-preservingly rewired ($10m$ edge swaps), Leiden is run on the rewired graph to obtain its fixed partition, and the identical pipeline is applied. Rewiring collapses modularity itself (base Q drops from 0.42–0.93 to 0.11–0.49, the residue a maximizer extracts from degree noise [9]), so any surviving signal is attributable to the degree sequence alone.

Table 4 shows the result: the null reproduces both signals on every network. DSpar separation is positive on 17/17 rewired graphs and exceeds the real graph's value on 16/17; fixed-partition modularity gains are positive on 17/17 and exceed the real graph's on 15/17, with median ratio 1.24. The sharpest cases are those where the real graph's signal

TABLE 4
Configuration-model null control. For each network the degree sequence is preserved exactly by double-edge swaps ($10m$ swaps), which destroys community structure; the sparsifier (DSpar, method="paper", nominal $\alpha = 0.8$) and the fixed-partition decomposition are then applied identically to both arms. The rewired null reproduces both signals commonly read as evidence of community structure: $\delta > 0$ in 17/17 nulls and $\Delta Q_{\text{fixed}} > 0$ in 17/17 nulls, with $\Delta Q_{\text{fixed}}(\text{null}) \geq \Delta Q_{\text{fixed}}(\text{real})$ in 15/17 networks (median ratio 1.24) and $\delta(\text{null}) \geq \delta(\text{real})$ in 16/17. facebook-combined is the only network with $\delta < 0$ on the real graph (bold), yet its null is still $\delta > 0$. Realized retention is m'/m: the with-replacement paper sampler retains far fewer distinct edges than the nominal α (see Table 1). Real arm: 1 Leiden partition × 3 sparsification seeds; null arm: 2 rewire seeds × 2 sparsification seeds (1 × 2 for cit-Patents and wiki-topcats). Standard deviations over replicates are shown for ΔQ_{fixed}; for the remaining columns they are ≤ 0.002 ($Q^{\text{base}}_{\text{fixed}}$), ≤ 0.004 (δ), ≤ 0.175 (hb) and ≤ 0.004 (retention).

		$Q^{\text{base}}_{\text{fixed}}$		δ		hb		ΔQ_{fixed}		realized ret.	
Network	m	real	null	real	null	real	null	real	null	real	null
email-Eu-core	16,064	0.4159	0.1274	+0.034	+0.041	2.113	1.404	+0.0819 ± 0.0044	+0.0350 ± 0.0032	0.444	0.447
wiki-Vote	100,736	0.4240	0.1260	+0.051	+0.101	1.902	1.689	+0.0711 ± 0.0012	+0.0843 ± 0.0012	0.332	0.333
ca-GrQc	13,422	0.8514	0.3768	+0.087	+0.189	0.478	1.758	+0.0176 ± 0.0058	+0.0991 ± 0.0031	0.452	0.474
ca-HepTh	24,806	0.7619	0.4129	+0.174	+0.215	1.581	1.929	+0.0523 ± 0.0022	+0.0941 ± 0.0016	0.477	0.483
facebook-combined	88,234	0.8355	0.1146	−0.001	+0.017	2.681	1.391	+0.0256 ± 0.0013	+0.0270 ± 0.0005	0.437	0.460
ca-CondMat	91,286	0.7312	0.3119	+0.113	+0.102	2.490	1.493	+0.0606 ± 0.0006	+0.0667 ± 0.0012	0.472	0.482
ca-HepPh	117,619	0.6512	0.1571	+0.021	+0.094	0.501	1.872	+0.0864 ± 0.0016	+0.0917 ± 0.0016	0.340	0.383
ca-AstroPh	196,972	0.6343	0.1676	+0.033	+0.041	1.496	1.294	+0.0393 ± 0.0008	+0.0489 ± 0.0011	0.417	0.436
email-Enron	180,811	0.6124	0.2420	+0.150	+0.240	2.974	4.472	+0.1422 ± 0.0004	+0.1562 ± 0.0013	0.389	0.394
cit-HepTh	352,021	0.6567	0.1583	+0.026	+0.032	2.173	1.778	+0.0461 ± 0.0002	+0.0376 ± 0.0006	0.434	0.453
cit-HepPh	420,784	0.7314	0.1675	+0.011	+0.026	1.358	1.300	+0.0126 ± 0.0003	+0.0319 ± 0.0004	0.459	0.468
com-Amazon	925,872	0.9296	0.4235	+0.010	+0.139	1.042	2.139	+0.0000 ± 0.0001	+0.0586 ± 0.0005	0.515	0.512
com-DBLP	1,049,866	0.8271	0.3647	+0.147	+0.197	1.583	2.308	+0.0437 ± 0.0002	+0.0980 ± 0.0005	0.469	0.475
com-Youtube	2,987,624	0.7188	0.4024	+0.246	+0.552	3.996	14.066	+0.0959 ± 0.0002	+0.2633 ± 0.0002	0.410	0.408
wiki-Talk	4,656,682	0.5995	0.4878	+0.513	+0.831	10.477	105.630	+0.2020 ± 0.0002	+0.3529 ± 0.0001	0.455	0.415
cit-Patents	16,511,740	0.8268	0.2962	+0.038	+0.166	0.723	1.940	+0.0047 ± 0.0001	+0.0768 ± 0.0000	0.461	0.467
wiki-topcats	25,444,207	0.6452	0.1412	+0.023	+0.029	5.076	6.920	+0.0484 ± 0.0001	+0.0538 ± 0.0000	0.447	0.435

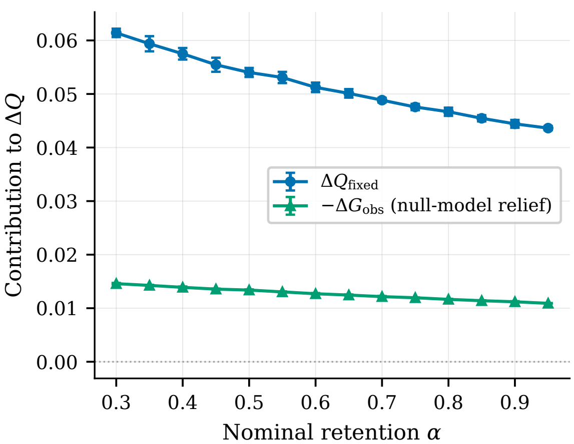

Fig. 2. Modularity decomposition under DSpar sparsification on cit-HepTh: ΔQ_{fixed} alongside the null-model contribution $-\Delta G_{\text{obs}}$ across sparsification levels; the remainder is ΔF_{obs} (Table 3).

is weakest: com-Amazon (real $\Delta Q_{\text{fixed}} = 1.5 \times 10^{-5}$, null 0.059), cit-Patents (0.005 vs. 0.077), and facebook-combined, whose real δ is negative (−0.0014) while its null is positive. Hub-bridging behaves identically: the ratio exceeds one on 17/17 nulls, reaching 105.6 on rewired wiki-Talk, while on the real graphs it is erratic (0.48–10.5), falling below one on three networks.

A degree-biased sampler preferentially thins hub-incident edges, which raises fixed-partition modularity for any partition on any degree-heterogeneous graph, whether or not the partition reflects communities. Positive δ, positive ΔQ_{fixed}, and hub-bridging above one are properties of heavy-tailed degree sequences interacting with modularity optimization; their sign carries no information about community structure. Consequently, correlations between δ and ΔQ_{fixed} across networks (however strong) validate only the arithmetic of Theorems 1–3, not any claim about community structure: the two quantities are linked mechanically through the same score distribution.

The control also yields a sharper observation: the signals are not merely reproduced but systematically stronger on the rewired graphs (δ larger on 16/17, ΔQ_{fixed} larger on 15/17). Real community structure evidently suppresses both quantities relative to pure degree noise, so the pair $(\delta, \delta_{\text{null}})$ does separate real networks from their configuration models, with the opposite sign from the naive reading; why real structure has this effect is an open question (Section 9).

Uniform random sparsification provides the complementary control: its fixed-partition change is zero within ±0.001 on real and rewired graphs alike, even at DSpar's realized retention, where DSpar registers +0.04 to +0.14; in expectation, unbiased edge deletion scales both modularity terms equally. The fixed-partition inflation therefore requires degree-biased removal specifically, while the cross-graph inflation of Section 6.1 requires only edge removal. The two artifacts are distinct mechanisms, and the two controls (configuration-model null; fixed-objective scoring) detect different failure modes: neither substitutes for the other.

5.3 Partition Provenance

A second confound concerns which partition δ is measured against. On standard LFR benchmarks [40] ($n = 10^4$, $\mu \in \{0.1, \ldots, 0.5\}$), δ is negative with respect to the planted partition at every mixing level (−0.003 to −0.010). Measured instead against a Leiden partition of the same graph, δ shifts systematically positive at every μ, and the shift grows exactly as Leiden departs from the planted truth (at $\mu = 0.5$: NMI to planted 0.67, planted communities merged from

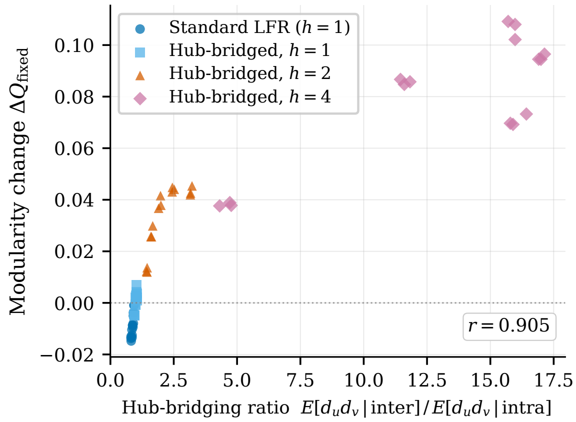

Fig. 3. Fixed-partition modularity change under DSpar as a function of induced hub-bridging on LFR graphs ($n = 10^4$, nominal $\alpha = 0.8$, planted partitions). The controlled rewiring operator (h: endpoint choice $\propto d^h$) restores the coupling between hub-bridging, δ, and $\Delta Q_{\mathrm{fixed}}$.

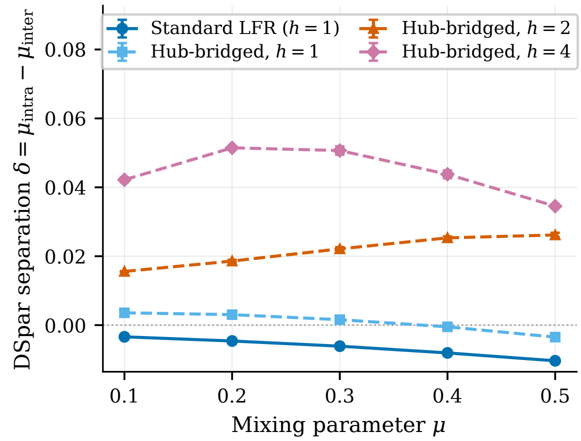

Fig. 4. DSpar separation δ (planted partitions) versus mixing parameter μ for standard ($h = 1$) and hub-bridged ($h > 1$) LFR graphs ($n = 10^4$).

236 to 48, shift $+0.0032$). The shift does not flip the sign on LFR, whose degree distribution is nearly homogeneous under standard parameterizations (coefficient of variation ≈ 0.4, versus 1.0–26 on the real networks in our suite), but on heavy-tailed real networks it can: on com-Amazon, $\delta = -0.004$ against the ground-truth communities and $+0.01$ against the Leiden partition of the same graph. Positive DSpar separation, as reported in practice, is therefore a joint property of heavy-tailed degrees and of partitions produced by modularity optimization; comparisons between δ values measured against planted partitions (synthetic benchmarks) and detected partitions (real networks) are not like-for-like.

5.4 Controlled Hub-Bridging in Synthetic Benchmarks

Finally, we include a controlled synthetic experiment, stated with the caveats above. Standard LFR graphs exhibit hub-bridging ratios below one and negative δ (planted partition), and sparsification lowers fixed-partition modularity ($\Delta Q_{\mathrm{fixed}} = -0.009$ at $\mu = 0.3$, $n = 10^4$, nominal $\alpha = 0.8$). We then rewire 70% of inter-community edges, re-attaching endpoints within their communities with probability proportional to d^h, where $h \geq 1$ controls the strength of degree preference; this preserves the number of inter-community edges (hence μ) while inducing controllable hub-bridging. At $h = 4$ the hub-bridging ratio reaches ≈ 15.9, δ turns positive ($+0.051$), and fixed-partition modularity change turns strongly positive ($+0.106$), with $\Delta Q_{\mathrm{fixed}}$ tightly coupled to the hub-bridging ratio across all configurations (Figure 3).

Read through the lens of §5.2, this experiment demonstrates controllability of the mechanism: degree–community coupling is the knob that turns fixed-partition gains on and off. It makes no statement about detection quality. It also identifies a genuine limitation of LFR as a benchmark for sparsification studies: standard LFR lacks both the heavy-tailed degree profile and the degree–community coupling that real networks and their configuration models exhibit.

6 Three Evaluation Artifacts

We now analyze the three evaluation designs by which sparsification appears to improve community detection, and pair each with the control that exposes it.

6.1 Artifact I: Modularity Is Not Comparable Across Graphs

The quantities $\Delta Q_{\mathrm{fixed}}$ and $\Delta Q^{(\mathrm{sp})}_{\mathrm{Leiden}}$ of Section 5 compare a partition's modularity on the sparsified graph against a (possibly different) partition's modularity on the original graph. Modularity, however, is defined relative to a graph's own configuration null model; changing the edge set changes the yardstick. Sparser graphs support higher modularity for combinatorial reasons unrelated to community structure [9], and altering m shifts the resolution at which modularity operates [26]. A quality claim requires a fixed objective: score every partition on the original graph.

Table 5 applies this control across 15 networks. We define the transfer loss as $Q_{\mathrm{base}} - Q_G(P_\alpha)$: the unsparsified baseline minus the transferred partition's modularity on the original graph. The partition obtained by running Leiden on the sparsified graph scores far above baseline on the sparsified graph (e.g., wiki-Talk: 0.897 vs. 0.602), and below baseline once evaluated on the original graph (0.417), a transfer loss of 0.185. Transfer loss is positive on all 15 networks at every tested retention, from 0.004 to 0.185. Cross-graph comparison thus reverses the sign of the conclusion on every dataset: what appears as a gain of $+0.01$ to $+0.30$ is, on the fixed objective, a loss of 0.004 to 0.19. Refining the transferred partition on the original graph (column Q_{final}) recovers the baseline almost exactly, leaving residuals of -0.001 to $+0.008$; Section 7 examines when those residuals are genuine. The artifact is not specific to DSpar: uniform random edge deletion, a sampler with no structural intelligence at all, reproduces the same sign flip on every configuration we tested (three networks, two retention levels: apparent gains of $+0.001$ to $+0.011$ on the sparsified graph, losses of 0.006–0.025 on the original). Cross-graph inflation is a property of the scoring convention, not of any particular sparsifier.

6.2 Artifact II: Partition Granularity

Sparsification fragments graphs, and detection on fragmented graphs returns finer partitions. Against small reference communities, as ground-truth sets in common benchmarks are,

TABLE 5
Artifact I — cross-graph modularity. Q_{sparse} is the modularity of the partition found on the sparsified graph, scored on that sparsified graph; Q_{orig} is the same partition scored on the original graph; the transfer loss is $Q_{\text{base}} - Q_{\text{orig}}$, the unsparsified baseline minus the transferred partition's score. It is positive for all 15 datasets at both α values, i.e. the headline "modularity gain" of the sparsify-then-cluster pipeline is an artifact of comparing scores computed on two different graphs. Q_{final} is Leiden seeded with the sparse-graph partition and run on the original graph, and ΔQ is that value minus the unsparsified baseline; the honest gain is ≤ 0.008 everywhere. Realized retention is m'/m (paper sampler, with replacement), not the nominal α. The $\alpha = 1.0$ rows of the source file are not baselines — the sampler still ran (realized retention 0.350–0.589) — and are excluded here; the unsparsified reference is the run's own Q_{base}, embedded in ΔQ. 5 seeds per cell; $\pm$ is the standard deviation over seeds (ΔQ and the transfer loss are reported without one, as the source file stores no standard deviation for them).

Dataset	α	realized ret.	Q_{sparse}	Q_{orig}	transfer loss	Q_{final}	ΔQ
ca-AstroPh	0.8	0.417	0.6902 ± 0.0011	0.6019 ± 0.0033	0.0334	0.6360 ± 0.0015	$+0.0006$
	0.9	0.446	0.6853 ± 0.0014	0.6047 ± 0.0035	0.0307	0.6358 ± 0.0012	$+0.0004$
ca-CondMat	0.8	0.473	0.8310 ± 0.0004	0.6597 ± 0.0017	0.0752	0.7348 ± 0.0013	-0.0000
	0.9	0.506	0.8234 ± 0.0006	0.6721 ± 0.0023	0.0628	0.7350 ± 0.0014	$+0.0001$
ca-GrQc	0.8	0.456	0.9002 ± 0.0003	0.7797 ± 0.0018	0.0722	0.8517 ± 0.0005	-0.0002
	0.9	0.485	0.8947 ± 0.0006	0.7924 ± 0.0024	0.0595	0.8518 ± 0.0005	-0.0001
ca-HepPh	0.8	0.339	0.7634 ± 0.0009	0.6209 ± 0.0056	0.0408	0.6610 ± 0.0008	-0.0006
	0.9	0.360	0.7605 ± 0.0010	0.6279 ± 0.0027	0.0338	0.6615 ± 0.0009	-0.0001
ca-HepTh	0.8	0.475	0.8825 ± 0.0005	0.6324 ± 0.0031	0.1320	0.7645 ± 0.0008	$+0.0002$
	0.9	0.507	0.8681 ± 0.0005	0.6532 ± 0.0031	0.1111	0.7644 ± 0.0009	$+0.0001$
cit-HepPh	0.8	0.460	0.7512 ± 0.0020	0.7220 ± 0.0039	0.0123	0.7349 ± 0.0009	$+0.0007$
	0.9	0.493	0.7486 ± 0.0022	0.7222 ± 0.0032	0.0121	0.7355 ± 0.0010	$+0.0012$
cit-HepTh	0.8	0.433	0.7178 ± 0.0013	0.6486 ± 0.0027	0.0143	0.6629 ± 0.0026	$+0.0000$
	0.9	0.465	0.7148 ± 0.0015	0.6509 ± 0.0018	0.0120	0.6624 ± 0.0026	-0.0005
com-Amazon	0.8	0.516	0.9519 ± 0.0001	0.8502 ± 0.0004	0.0815	0.9324 ± 0.0001	$+0.0007$
	0.9	0.554	0.9480 ± 0.0001	0.8728 ± 0.0004	0.0589	0.9325 ± 0.0001	$+0.0008$
com-DBLP	0.8	0.469	0.9090 ± 0.0003	0.7429 ± 0.0018	0.0879	0.8338 ± 0.0004	$+0.0031$
	0.9	0.501	0.9024 ± 0.0002	0.7599 ± 0.0011	0.0709	0.8337 ± 0.0004	$+0.0029$
com-Youtube	0.8	0.410	0.8901 ± 0.0005	0.5483 ± 0.0043	0.1801	0.7281 ± 0.0002	-0.0003
	0.9	0.434	0.8789 ± 0.0002	0.5811 ± 0.0019	0.1472	0.7281 ± 0.0003	-0.0003
email-Enron	0.8	0.389	0.7942 ± 0.0006	0.5683 ± 0.0026	0.0464	0.6219 ± 0.0021	$+0.0072$
	0.9	0.413	0.7860 ± 0.0009	0.5739 ± 0.0031	0.0408	0.6225 ± 0.0017	$+0.0078$
facebook-combined	0.8	0.436	0.8625 ± 0.0003	0.8314 ± 0.0009	0.0043	0.8357 ± 0.0001	$+0.0000$
	0.9	0.467	0.8624 ± 0.0003	0.8313 ± 0.0009	0.0044	0.8355 ± 0.0005	-0.0002
soc-Epinions1	0.8	0.317	0.6931 ± 0.0021	0.3225 ± 0.0115	0.1283	0.4500 ± 0.0033	-0.0008
	0.9	0.334	0.6783 ± 0.0019	0.3439 ± 0.0114	0.1068	0.4514 ± 0.0034	$+0.0006$
wiki-Talk	0.8	0.455	0.8974 ± 0.0000	0.4169 ± 0.0004	0.1852	0.6027 ± 0.0001	$+0.0006$
	0.9	0.483	0.8860 ± 0.0000	0.4373 ± 0.0004	0.1648	0.6028 ± 0.0002	$+0.0007$
wiki-Vote	0.8	0.333	0.5119 ± 0.0055	0.4004 ± 0.0065	0.0244	0.4250 ± 0.0033	$+0.0002$
	0.9	0.357	0.5104 ± 0.0039	0.4081 ± 0.0027	0.0168	0.4255 ± 0.0034	$+0.0007$

finer partitions score better on popular agreement measures for reasons unrelated to the preprocessing: uncorrected NMI is biased toward many clusters [10], and set-matching scores such as average F_1 reward granularity against small targets, since splitting clusters moves candidate sets toward the size of the small reference communities. The control is to grant the baseline the same granularity: run the same algorithm on the unsparsified graph with the resolution parameter tuned to match the sparsified pipeline's cluster count.

6.2.0.1 At scale.: On SNAP networks with functional ground truth (top-5000 communities; median size 7–8 nodes), the with-replacement DSpar pipeline at nominal $\alpha = 0.8$ appears to help on two of three datasets (average F_1, mean of both matching directions: com-Amazon $0.211 \to 0.285$; com-Youtube $0.010 \to 0.050$; com-DBLP $0.146 \to 0.100$), while inflating cluster counts 21–42×. The resolution-matched control, using every edge, beats the sparsified pipeline everywhere, by wide margins: 0.404 on com-Amazon, 0.293 on com-DBLP, 0.129 on com-Youtube. The apparent recovery gain is obtainable, doubled, without discarding a single edge; the discarded edges are pure loss. Mild pruning via the clipped no-replacement sampler at nominal 0.9 (true retention 0.52–0.76) moves recovery by -0.037 to $+0.001$, indistinguishable from zero.

6.2.0.2 On classical labeled graphs.: Table 6 repeats the analysis on five classical labeled datasets, adding chance-corrected AMI [10] as the primary measure. Four of the five datasets are negative under every metric, and become more negative under AMI (e.g., Karate $\Delta\text{AMI} = -0.19$). The one apparent positive, email-Eu-core, does not survive: its $\Delta\text{AMI} = +0.007 \pm 0.019$ is statistically indistinguishable from zero, and the resolution-matched control exceeds the sparsified pipeline on all three metrics. Under calibrated true-0.9 retention the effect is negative even nominally; the residual positive depended on the with-replacement sampler discarding 56% of edges, not on the degree scores. We note additionally that reference labels are themselves imperfect proxies for structural communities [27], [28], and that one dataset's conventional labels proved on inspection to be an algorithmic surrogate rather than observed metadata (Table 6, footnote).

6.3 Artifact III: Unmatched Computation

A sparsify–cluster–refine pipeline performs more optimization than the single baseline run it is typically compared against. The control is to grant the baseline equal wall-clock time,

TABLE 6
Artifact II — granularity. Ground-truth recovery on the five small benchmark graphs, 10 seeds per cell, mean ± standard deviation. baseline is Leiden on the original graph; DSpar paper is the with-replacement pipeline (nominal $\alpha = 0.8$, realized retention 0.44–0.55); the resolution-matched control applies no sparsification at all and instead tunes the RBConfiguration resolution γ on the original graph until the cluster count matches the DSpar run; DSpar calibrated uses the λ-bisection sampler with $\mathbb{E}[\text{retention}] = 0.9$ exactly. AMI is the chance-corrected primary metric; NMI is commonly reported in this literature and is inflated by the finer partitions sparsification produces. The one apparently positive dataset (email-Eu-core) is reproduced and exceeded by the resolution-matched control at 100% of the edges (ΔAMI +0.014 vs +0.007), so the score against a granularity-matched baseline is 0/5.

Dataset	Condition	true ret.	γ	#clusters	AMI	ARI	NMI
Karate	baseline (no sparsification)	1.000	1.000	4.0 ± 0.0	0.557 ± 0.028	0.457 ± 0.022	0.579 ± 0.027
	DSpar paper, $\alpha = 0.8$	0.541	1.000	7.3 ± 1.7	0.366 ± 0.059	0.294 ± 0.073	0.429 ± 0.050
	resolution-matched control	1.000	2.000	7.3 ± 0.5	0.392 ± 0.005	0.239 ± 0.009	0.445 ± 0.003
	DSpar calibrated, $\alpha = 0.9$	0.887	1.000	4.0 ± 0.0	0.492 ± 0.038	0.416 ± 0.031	0.517 ± 0.036
Dolphins†	baseline (no sparsification)	1.000	1.000	5.0 ± 0.4	0.270 ± 0.029	0.246 ± 0.033	0.293 ± 0.027
	DSpar paper, $\alpha = 0.8$	0.520	1.000	9.7 ± 1.2	0.216 ± 0.045	0.132 ± 0.029	0.264 ± 0.039
	resolution-matched control	1.000	2.300	10.3 ± 0.9	0.223 ± 0.018	0.125 ± 0.010	0.270 ± 0.014
	DSpar calibrated, $\alpha = 0.9$	0.892	1.000	4.9 ± 0.5	0.261 ± 0.021	0.195 ± 0.047	0.284 ± 0.019
Football	baseline (no sparsification)	1.000	1.000	9.7 ± 0.5	0.849 ± 0.018	0.779 ± 0.047	0.880 ± 0.016
	DSpar paper, $\alpha = 0.8$	0.549	1.000	9.2 ± 0.9	0.812 ± 0.032	0.737 ± 0.059	0.849 ± 0.028
	resolution-matched control	1.000	1.000	9.7 ± 0.5	0.849 ± 0.018	0.779 ± 0.047	0.880 ± 0.016
	DSpar calibrated, $\alpha = 0.9$	0.899	1.000	8.9 ± 0.3	0.816 ± 0.013	0.702 ± 0.027	0.852 ± 0.011
Polbooks	baseline (no sparsification)	1.000	1.000	4.7 ± 0.5	0.535 ± 0.026	0.625 ± 0.061	0.551 ± 0.024
	DSpar paper, $\alpha = 0.8$	0.527	1.000	6.8 ± 0.7	0.413 ± 0.043	0.341 ± 0.069	0.439 ± 0.041
	resolution-matched control	1.000	1.500	6.8 ± 0.6	0.431 ± 0.012	0.333 ± 0.021	0.456 ± 0.010
	DSpar calibrated, $\alpha = 0.9$	0.900	1.000	5.2 ± 0.7	0.475 ± 0.028	0.490 ± 0.088	0.494 ± 0.025
email-Eu-core	baseline (no sparsification)	1.000	1.000	7.7 ± 0.5	0.558 ± 0.010	0.323 ± 0.021	0.583 ± 0.011
	DSpar paper, $\alpha = 0.8$	0.443	1.000	8.6 ± 0.5	0.565 ± 0.014	0.360 ± 0.026	0.592 ± 0.013
	resolution-matched control	1.000	1.075	8.6 ± 0.9	0.571 ± 0.017	0.361 ± 0.043	0.598 ± 0.018
	DSpar calibrated, $\alpha = 0.9$	0.900	1.000	7.3 ± 0.6	0.549 ± 0.024	0.298 ± 0.037	0.573 ± 0.024

† Dolphins has no canonical ground-truth partition; the labels used here are a surrogate two-group split obtained by Kernighan–Lin bisection, so its recovery scores measure agreement with a heuristic cut, not with metadata.

spent on independent restarts, and compare against the best. In our measurements the pipeline costs 1.5–2.0× a single Leiden run (sparsification itself is 2–10% of the total), so the matched baseline is best-of-two restarts. Best-of-two costs 2.0× a single run, at or slightly above the pipeline's cost; the slack favors the baseline, which is conservative for reported gains and mildly anti-conservative for reported losses. The correction is not cosmetic: on wiki-Vote, the seeded pipeline beats the mean baseline in six of eight configurations and loses to the runtime-matched baseline in all eight. Comparisons against single-run means manufacture false positives whenever the optimizer's seed variance is nonzero.

6.4 The Resulting Protocol

The three controls are general and cheap, and we suggest them as a minimal standard for any claim that preprocessing improves community detection:

1) Fixed objective: evaluate all partitions on the original graph (or an external reference), never on the preprocessed one.
2) Granularity matching: give the baseline the preprocessing pipeline's cluster count via its resolution parameter; use chance-corrected agreement measures.
3) Compute matching: give the baseline the pipeline's wall-clock budget via restarts, and compare best against best.

To certify that a structural signal reflects community structure rather than degree structure, add the configuration-model null of Section 5.2.

7 The Boundary: Where Genuine Gains Exist

Having removed the artifacts, we ask what remains. The one effect that survives all three controls is seeded refinement: run Leiden on the sparsified graph, transfer the partition to the original graph, and continue optimizing there. This uses sparsification not as a replacement for the graph but as a perturbation of the optimizer's search trajectory; the final quantity is modularity on the original graph, so Artifact I does not arise, and we compare against runtime-matched restarts, so Artifact III does not arise.

7.1 Runtime-Matched Seeded Refinement

Table 7 reports the outcome on 15 networks at calibrated retention 0.9. In the screening experiment, the seeded pipeline beats the runtime-matched baseline beyond twice the baseline's seed deviation on exactly 2 of 15 networks: email-Enron (+0.0077, 4.1 baseline standard deviations) and com-Youtube (+0.0086, 2.5); across 15 networks at a two-deviation screening threshold, roughly half a false positive is expected. The com-Youtube margin as screened is inflated: its matched baseline is a single best-of-two realization that landed at the second percentile of the exact best-of-two distribution (all 45 pairs over ten baseline runs). A dedicated sweep (both samplers, $\alpha \in \{0.7, 0.8, 0.9, 0.95\}$, five seeds each) gives the corrected picture. Against the expected best-of-two, the gain is confined to the calibrated sampler at $\alpha \geq 0.9$ and is small (+0.0023 at $\alpha = 0.9$, +0.0031 at $\alpha = 0.95$); the clipped sampler is negative in all four of its configurations, and aggressive calibrated sparsification ($\alpha = 0.7$) is strongly negative. The $\alpha = 0.95$ cell is nonetheless statistically solid: its seeded mean beats every one of the 45 matched best-of-two draws

TABLE 7
The boundary: runtime-matched seeded refinement, 15 networks, calibrated sampler at $\alpha = 0.9$ (realized retention ≈ 0.900 by construction). Q_{base} is plain Leiden on the original graph (5 seeds); Q_{seeded} is Leiden on the original graph seeded with the sparse-graph partition (3 sparsification seeds); Q_{matched} is the best of the plain-Leiden restarts affordable within the same wall-clock budget as the full sparsify + cluster + seed pipeline (2 restarts in all 15 cells). The honest outcome is $y = Q_{\text{seeded}} - Q_{\text{matched}}$. Only 2/15 networks clear $y > 2\sigma_{\text{base}}$ (starred, bold): email-Enron (robust across all tested configurations) and com-Youtube (small gain confined to mild calibrated retention; see text). Genuine gains therefore exist, but they are rare and small (≤ 0.009).

Network	n	m	ret.	Q_{base}	Q_{seeded}	Q_{matched}	y	$y > 2\sigma_{\text{base}}$
email-Eu-core	986	16,064	0.900	0.4160 ± 0.0004	0.4151 ± 0.0021	0.4174	-0.0023	no
wiki-Vote	7,066	100,736	0.900	0.4244 ± 0.0025	0.4234 ± 0.0017	0.4294	-0.0059	no
ca-GrQc	4,158	13,422	0.900	0.8506 ± 0.0010	0.8521 ± 0.0002	0.8517	$+0.0004$	no
ca-HepTh	8,638	24,806	0.901	0.7614 ± 0.0009	0.7605 ± 0.0010	0.7621	-0.0016	no
facebook-combined	4,039	88,234	0.900	0.8356 ± 0.0001	0.8356 ± 0.0003	0.8355	$+0.0001$	no
ca-CondMat	21,363	91,286	0.900	0.7308 ± 0.0011	0.7312 ± 0.0005	0.7294	$+0.0018$	no
ca-HepPh	11,204	117,619	0.900	0.6583 ± 0.0010	0.6599 ± 0.0016	0.6602	-0.0003	no
ca-AstroPh	17,903	196,972	0.900	0.6308 ± 0.0027	0.6328 ± 0.0007	0.6316	$+0.0013$	no
email-Enron	33,696	180,811	0.900	0.6071 ± 0.0019	0.6128 ± 0.0014	0.6051	$+0.0077^*$	yes
cit-HepTh	27,400	352,021	0.900	0.6621 ± 0.0023	0.6611 ± 0.0018	0.6579	$+0.0032$	no
cit-HepPh	34,401	420,784	0.900	0.7319 ± 0.0019	0.7335 ± 0.0014	0.7343	-0.0008	no
com-Amazon	334,863	925,872	0.900	0.9294 ± 0.0003	0.9300 ± 0.0001	0.9296	$+0.0004$	no
com-DBLP	317,080	1,049,866	0.900	0.8257 ± 0.0008	0.8281 ± 0.0002	0.8275	$+0.0006$	no
com-Youtube	1,134,890	2,987,624	0.900	0.7245 ± 0.0034	0.7296 ± 0.0001	0.7209	$+0.0086^*$	yes
wiki-Talk	2,388,953	4,656,682	0.900	0.6011 ± 0.0007	0.6015 ± 0.0003	0.6020	-0.0006	no

and all ten individual baseline runs (bootstrap $p < 10^{-4}$, surviving Bonferroni correction across the eight configurations), with across-seed spread ± 0.0004. That the gain rises as the sparsifier removes less marks it as a perturbation-of-the-optimizer effect rather than a sparsification benefit. We therefore report one substantial robust gain (email-Enron) and one small, regime-confined gain (com-Youtube). The email-Enron result is robust across the sweep: against the runtime-matched best-of-two baseline it is positive in all eight configurations (both sampler variants, every retention level 0.7–0.95; margins +0.0007 to +0.0149, the largest under the clipped no-replacement sampler at nominal $\alpha = 0.7$, true retention 0.45), and against the stricter best-of-five baseline it is positive in seven of eight (margins up to +0.0107; the exception is calibrated $\alpha = 0.7$ at -0.004). That the margins grow under harsher sparsification indicates the gain stems from aggressive hub-edge removal changing the optimizer's search space, not from mild denoising. On email-Enron the raw transferred partition itself (before refinement) already exceeds baseline at true retention ≥ 0.8, the only network in the suite where the transfer loss of Section 6.1 changes sign.

7.2 No Tested Statistic Predicts the Gains

Can one tell in advance where seeding will pay? Table 8 correlates fourteen candidate statistics with the runtime-matched outcome: δ, its configuration-null value and excess $\delta^* = \delta_{\text{real}} - \delta_{\text{null}}$, the fixed-gain ratio against the null, hub-bridging and its excess, degree heterogeneity, size, and density. None is significant at $n = 15$ (best: network size, Spearman $\rho = 0.47$, $p = 0.079$; the significance threshold after correcting for fourteen candidates is $|\rho| \geq 0.70$). The null-corrected excess δ^*, superficially the principled choice, is in fact anti-correlated with the outcome ($r = -0.36$): because δ_{null} exceeds δ_{real} on 16 of 17 networks and tracks degree heterogeneity, δ^* behaves as a negated heavy-tail statistic. Its proper role is diagnostic, quantifying how much of a reported δ the configuration model already explains; it is not predictive.

One near-miss is worth recording as a methodological caution. Over the first fourteen networks measured, degree heterogeneity predicted the outcome at conventional significance ($r = 0.69$, $p = 0.006$). The fifteenth, wiki-Talk, the most degree-heterogeneous network in the suite (CV = 26.3, hub-bridging 10.5), returned a negative outcome and collapsed the correlation to $r = 0.12$ ($p = 0.66$). A single network eliminated an apparently significant fourteen-point correlation. The earlier version of this work reported a cross-network correlation of $r = 0.92$ between δ and fixed-partition gains as a headline validation; beyond being mechanically induced (Section 5.2), such small-sample correlations are fragile in exactly the way exhibited here. Predictor claims in this regime require samples of order 10^2 networks and pre-registered statistics.

7.3 Retention Dependence

Under calibrated sampling, the transfer loss of Artifact I is governed by true retention: it shrinks roughly tenfold from retention 0.7 to 0.95 (e.g., ca-HepTh: $-0.037 \rightarrow -0.0035$; com-DBLP: $-0.027 \rightarrow -0.0010$) and extrapolates to zero only at retention one. On five of six networks examined across the sweep, no retention level yields a positive raw transfer; email-Enron again is the exception, crossing between 0.7 and 0.8. On the fixed objective, milder sparsification is simply a smaller loss, vanishing only when nothing is removed; the exception is the seeding effect above, which no tested statistic anticipates.

8 Runtime Considerations

Sparsification's traditional justification is speed, so we report the runtime picture honestly; it is less favorable than commonly assumed.

8.0.0.1 Clustering time does not track edge count.: At realized retention near one half (nominal $\alpha = 0.8$, with-replacement sampler), Leiden on the sparsified graph was slower than on the original on six of eight large networks in our measurements, spanning up to 117M edges (speedup 0.48–0.89); the exceptions were wiki-topcats (1.26) and,

TABLE 8
No predictor of the genuine gain. Correlations between 14 candidate structural statistics and the seeded-refinement outcome across the $n = 15$ networks of Table 7. $y = Q_{\text{seeded}} - Q_{\text{matched}}$ is the runtime-matched (honest) outcome; $y_2 = Q_{\text{seeded}} - Q_{\text{base}}$ is the naive one. No predictor is significant for y at the 5% level; the strongest is n ($\rho = 0.468$, $p = 0.079$). Two cells reach nominal significance against the naive outcome only (hb excess, $p = 0.026$; n, $p = 0.044$), and neither survives correction for the 28 tests reported here. δ^* — the null-corrected separation statistic — is wrong-signed for both outcomes and is therefore retained as a diagnostic only. With $n = 15$, the 95% CI of a single Pearson r spans roughly ± 0.5; see the wiki-Talk case study in the text, where dropping one network moves r(deg CV) from 0.69 to 0.12.

	y (runtime-matched)				y_2 (naive)			
Predictor	ρ	p	r	p	ρ	p	r	p
δ (real graph)	+0.171	0.541	+0.207	0.460	+0.196	0.483	+0.205	0.464
$\delta^* = \delta_{\text{real}} - \delta_{\text{null}}$	−0.075	0.791	−0.360	0.188	−0.386	0.156	−0.398	0.142
$\delta_{\text{real}}/\delta_{\text{null}}$	+0.104	0.713	+0.059	0.834	−0.246	0.376	−0.091	0.747
$\Delta Q^{\text{real}}_{\text{fixed}}/\Delta Q^{\text{null}}_{\text{fixed}}$	−0.089	0.752	−0.166	0.556	−0.429	0.111	−0.341	0.214
$\Delta Q^{\text{real}}_{\text{fixed}} - \Delta Q^{\text{null}}_{\text{fixed}}$	−0.146	0.603	−0.346	0.207	−0.450	0.092	−0.401	0.138
hb (real graph)	+0.304	0.271	+0.129	0.646	−0.050	0.860	+0.036	0.899
$\text{hb}_{\text{real}} - \text{hb}_{\text{null}}$	−0.211	0.451	+0.038	0.893	−0.571	0.026	+0.040	0.888
$\text{hb}_{\text{real}}/\text{hb}_{\text{null}}$	−0.050	0.860	−0.234	0.400	−0.436	0.104	−0.445	0.097
degree CV $\sigma_d/\bar{d}$	+0.339	0.216	+0.122	0.664	+0.346	0.206	+0.109	0.699
$Q^{\text{base}}_{\text{fixed}}$ (real graph)	+0.200	0.475	+0.257	0.355	+0.211	0.451	+0.179	0.522
$Q^{\text{base,real}}_{\text{fixed}}/Q^{\text{base,null}}_{\text{fixed}}$	−0.118	0.676	−0.185	0.510	−0.150	0.594	−0.242	0.384
average degree $\bar{d}$	−0.286	0.302	−0.358	0.191	−0.396	0.143	−0.403	0.137
n (nodes)	+0.468	0.079	+0.164	0.558	+0.525	0.044	+0.144	0.609
m (edges)	+0.375	0.168	+0.248	0.374	+0.450	0.092	+0.224	0.423

marginally, com-Orkut (1.00). The cause is structural: sparsification fragments the graph and inflates the number of detected communities (by factors of 21–42 in the ground-truth experiments of Section 6.2), and Leiden's per-community bookkeeping grows accordingly. At aggressive retention the fragmentation regime dominates and runtime increases despite the smaller edge set. Speedup claims should therefore be read jointly with fragmentation diagnostics (component counts, largest-component fraction, community counts), and the fragmented regime treated as out of scope for quality claims of any kind. Timing scripts and per-run logs for the measurements quoted in this section are included in the supplementary repository.

8.0.0.2 Pipeline accounting.: DSpar scoring itself is cheap (2–10% of the seeded-refinement pipeline's time in our measurements, consistent with its $O(m)$ design), so the pipeline cost is dominated by clustering, and the seeded-refinement pipeline of Section 7 costs 1.5–2.0× a single baseline run. This is exactly why Artifact III matters: the honest comparison for any such pipeline is against a baseline granted the same budget.

8.0.0.3 Spectral baselines.: We do not report effective-resistance sparsification comparisons. In auditing the earlier version of this work we found its spectral branch had silently returned the original graph essentially unmodified (identical or near-identical edge counts, 99.8–100%, and clustering times). The cause is instructive: the map from the retention parameter to the sparsifier's ϵ was miscalibrated by roughly two orders of magnitude, leaving the effective-resistance keep-probabilities, which scale as $1/\epsilon^2$ and clip at one, saturated for four of its five settings; the reweighting required by the spectral guarantee was additionally discarded. Nothing in the pipeline verified realized retention, so the defect was invisible in every quality metric. This failure mode is precisely why we recommend logging realized retention as part of the protocol; a corrected spectral comparison is left to future work.

9 Discussion and Conclusion

9.0.0.1 What the theory licenses.: Our fixed-partition results (Theorems 1–3) are correct and verified to high precision, and they should be read for exactly what they state: degree-biased sampling shifts edge composition and null-model mass in a direction determined by the score gap δ, for whatever partition δ is measured against. The configuration-model controls show that this premise, and therefore the conclusion, holds as readily on graphs without communities as on real networks. The theory is a quantitative account of degree mechanics; treating it as a theory of community clarification was an over-interpretation. We made this over-interpretation ourselves, and the literature on preprocessing for clustering is structurally exposed to it: the certifying statistics look meaningful, and the standard evaluations all move in the flattering direction.

9.0.0.2 What we recommend.: For practitioners: degree-based sparsification remains a legitimate scaling tool at mild retention, where the fixed-objective cost is small (Section 7.3), and seeded refinement is worth trying when clustering budgets allow a second pass, but no quality improvement should be assumed, and none should be claimed without the controls of Section 6.4. For evaluators and reviewers: the three controls (fixed objective, granularity matching, compute matching) plus the configuration-model null are each a few lines of code, and each reversed at least one published-style conclusion in this paper. For benchmark designers: standard LFR is doubly unrepresentative for sparsification studies, being nearly homogeneous in degree and uncoupled between degree and community membership; controlled degree–community coupling of the kind used here (or in richer generators) is required for synthetic conclusions to transfer.

9.0.0.3 Limitations.: Our scope is modularity-based detection with Leiden under DSpar-family sparsification; other objectives (description-length, flow-based), other detectors, and similarity- or resistance-based sparsifiers may behave differently, and our protocol is designed to test exactly that. The boundary characterization rests on 15 networks: sufficient to establish existence (2 genuine successes) and to defeat the tested predictors, insufficient to establish what does predict success. Reference-label experiments inherit the known gap between metadata and structural communities [27], [28].

9.0.0.4 Open problems.: Why email-Enron and com-Youtube? Both gains are real and robust, yet none of the statistics we computed forecasts them. What property of these two graphs lets hub-edge removal steer the optimizer to better optima is the concrete open question this work leaves. A second is empirical: why does real community structure systematically suppress δ and the fixed-partition gain relative to the configuration null (Section 5.2)? A third is constructive: whether a sparsifier can be designed so that its selection signal separates real networks from their configuration models, rather than inheriting the degree sequence's behavior.

9.0.0.5 Conclusion.: We set out to determine when sparsification helps community detection. The answer has three parts. The mechanism by which degree-based sparsification reshapes graphs is real, provable, and controllable. The standard evidence that this mechanism improves detection is artifactual: it is attributable to cross-graph scoring, granularity, and unmatched computation, and it is reproduced in full by configuration models without community structure. And a genuine benefit exists but is rare and currently unpredictable: two networks in fifteen, small margins, surviving every control we could construct. Sparsification is a scaling tool that in rare and so far unpredictable cases also serves as a useful optimizer perturbation; it is not a general clarifier of community structure. The evaluation protocol that separates these outcomes costs little, and we hope it becomes routine.

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Appendix

Reproducibility

All experiments in this paper are reproducible from the accompanying repository: per-experiment scripts, per-run CSV outputs, fixed seeds, and the calibrated sampler implementation, together with the audit artifacts (including bit-level verification of our reimplementation of the with-replacement sampler against the reference implementation). Realized retention is logged for every sparsification call. Dataset sources: SNAP collections for the large networks and standard repositories for the classical labeled graphs, as cited per table.

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